

Presentation

Comparative Analysis and Performance Evaluation of a Two -Wheeled Self -Balancing Robot Using P, PI, and PID Controllers Tuned By Metaheuristic Algorithms

ME 461

Control Engineering

**Bangladesh University of
Engineering and Technology**

Group-05

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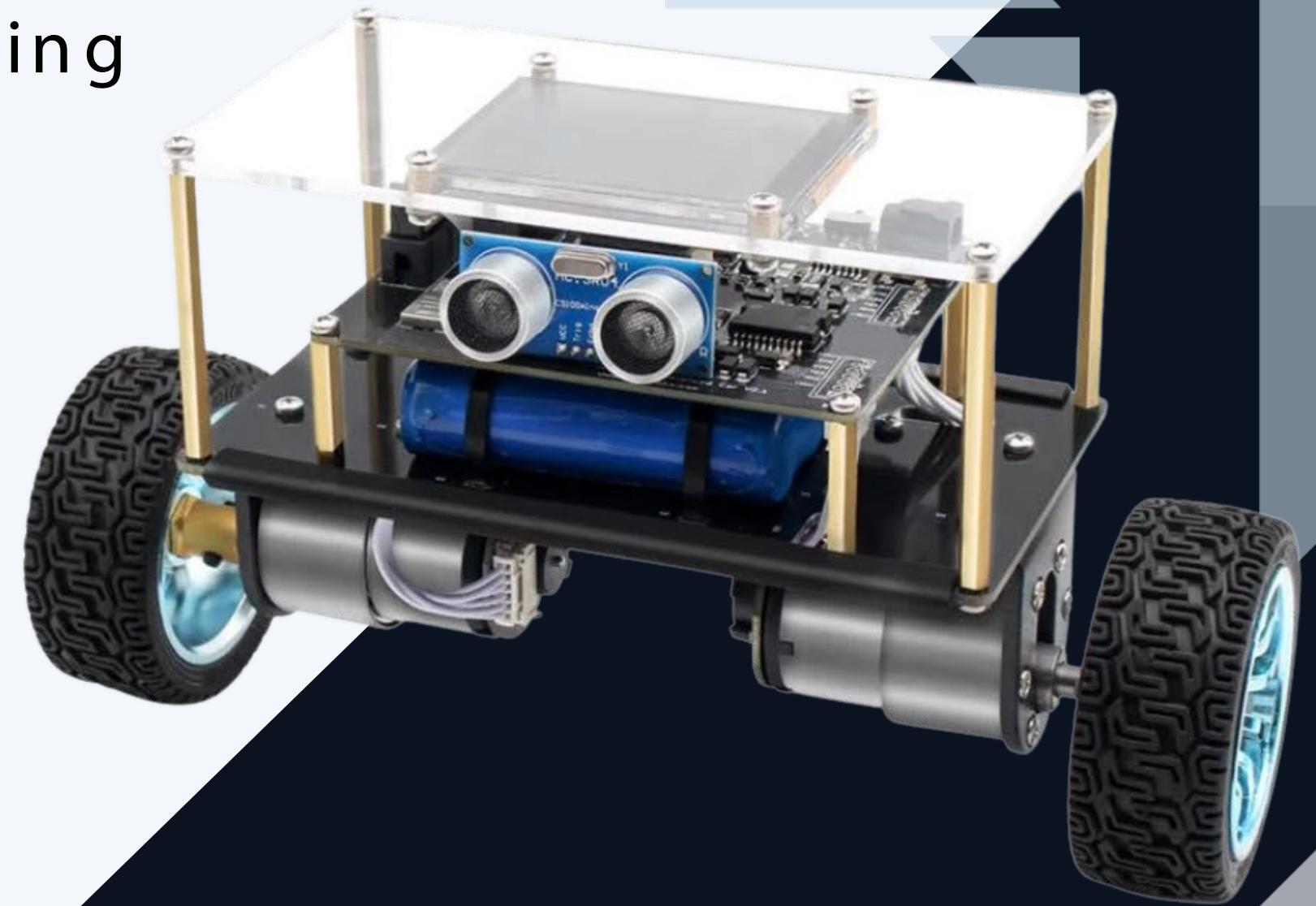
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Introduction

What is a two-wheeled self-balancing robot?

1. A dynamically unstable mobile system modeled as an inverted pendulum on wheels .
2. Requires continuous feedback control to maintain balance and follow commands .





Literature Review

Author Name	Modeling Approach	Control Technique	Trajectory Planning	Hardware Implementation	Linearized Model
Mahler et al. (2013)	Lagrangian + Full Dynamics	Full State Feedback + PI	None	Real Prototype	Yes (State-space from linearized model)
Frankovský et al. (2017)	Lagrangian + Gearmotor	State Feedback (Linear)	None	Simulation only	Yes (Linearized from Lagrangian model)
Le et al. (2024)	Euler–Lagrange + Constraints	HSMC + SMC + Kinematic Controller	RRT-based planning	Simulations (MATLAB)	No (Nonlinear model used)
Halawa et al. (2023)	Basic Transfer Function	PID	None	Physical Robot Built	Simplified Transfer Function



Motivation

1. Mahler et al. (2013) developed a mathematical model and control strategy using full θ -state feedback and PI control.
2. However, their work lacked comparative analysis of simpler PID θ -based approaches using different controlling algorithms.





Our Contribution

1. Evaluate P, PI, and PID controllers using metaheuristic algorithms for gain optimization.
2. Assess performance based on transient and steady -state criteria.
3. Provide insights into how intelligent optimization improves classical control.





Mathematical Modelling

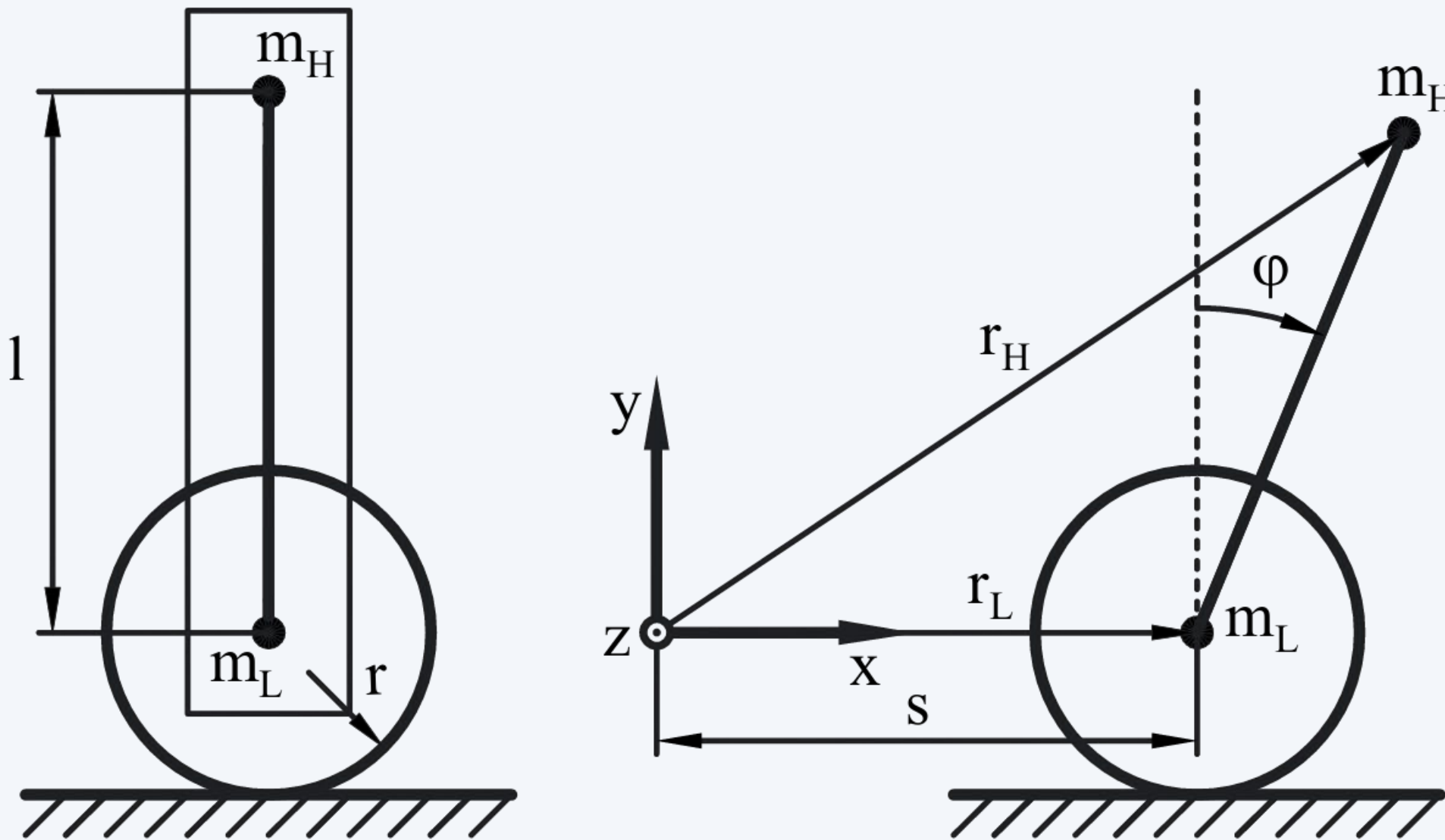


Fig: Structure and definitions of the mathematical model



Mathematical Modelling

$$\mathbf{r}_H = [s + l \sin(\varphi), 0, l \cos(\varphi)]^T$$

$$\mathbf{r}_L = [s, 0, 0]^T$$

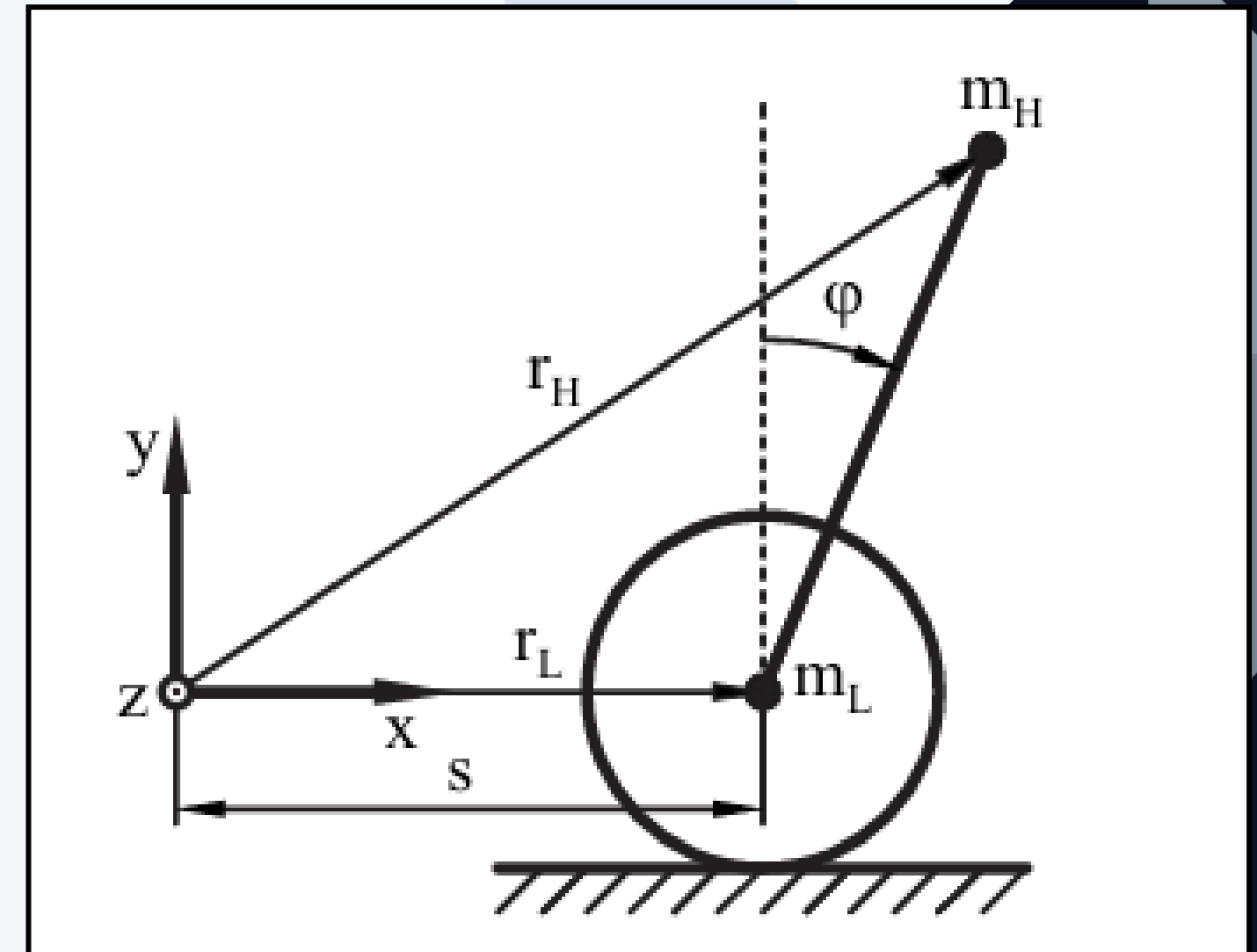
$$\mathbf{v} = \dot{\mathbf{s}}$$

$$\omega = \dot{\varphi}$$

The Lagrangian of the system is defined as:

$$\mathcal{L} = T - V$$

where T is the total kinetic energy and V is the potential energy of the system.





Mathematical Modelling

The kinetic energy is composed of:

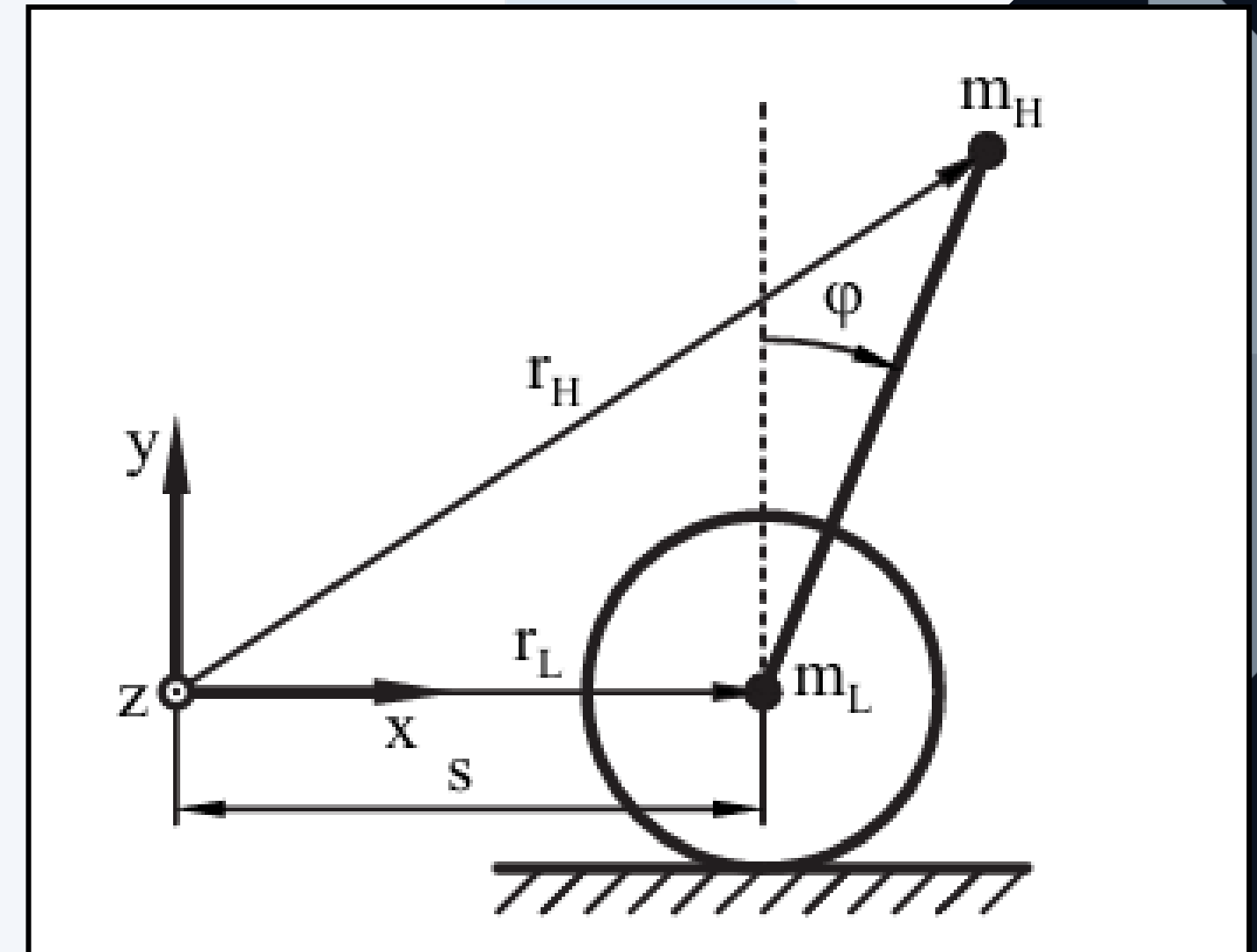
$$T_H = \frac{1}{2} m_H (\dot{s}^2 + 2l\dot{s}\dot{\phi} \cos \phi + l^2 \dot{\phi}^2)$$

$$T_L = \frac{1}{2} m_L \dot{s}^2$$

$$T_R = \frac{1}{2} J_R \left(\frac{\dot{s}}{r} \right)^2$$

The potential energy is:

$$V = m_H g l \cos \phi$$





Mathematical Modelling

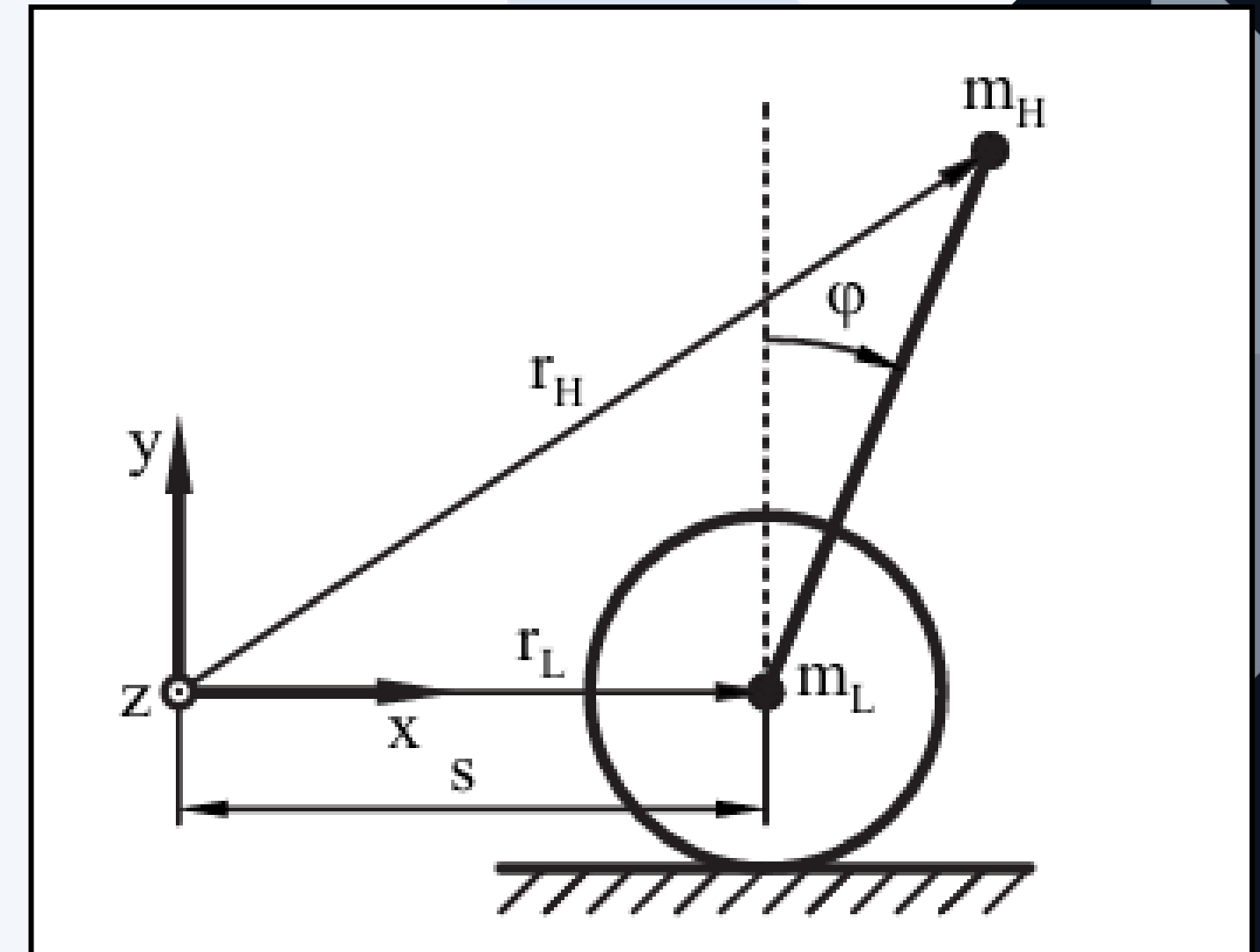
The dissipative effects are modeled using Rayleigh's dissipation function:

$$D = \frac{1}{2} \left(c_v \dot{s}^2 + c_\omega \left(\frac{\dot{s}}{r} - \dot{\phi} \right)^2 \right)$$

Applying the Euler-Lagrange equation with dissipation:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} + \frac{\partial D}{\partial \dot{q}_i} = \tau_i$$

$$\begin{aligned} \frac{d\omega}{dt} = & \frac{((m_H + m_L)r^2 + J_R)(g - \omega v) \sin(\varphi)}{l((m_H + m_L)r^2 + J_R - m_H r^2 \cos^2(\varphi))} \\ & + \frac{((M_D + \omega c_\omega)r - c_v v r^2 - c_\omega v) \cos(\varphi)}{l((m_H + m_L)r^2 + J_R - m_H r^2 \cos^2(\varphi))} \\ & - \frac{((M_D + \omega c_\omega)r - c_\omega v)((m_H + m_L)r^2 + J_R)}{m_H l^2 r((m_H + m_L)r^2 + J_R - m_H r^2 \cos^2(\varphi))} \end{aligned}$$





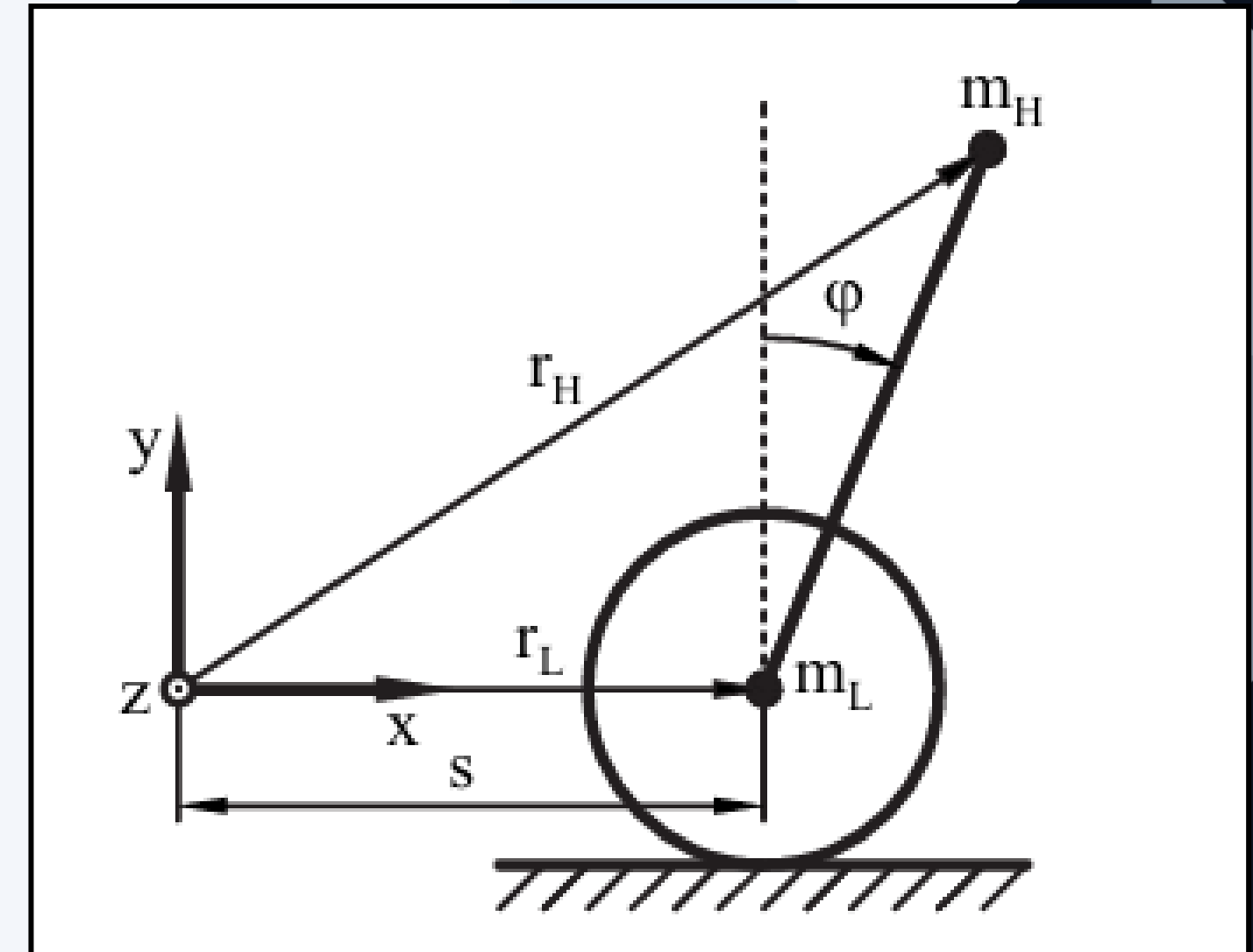
Mathematical Modelling

$$\frac{dv}{dt} = \frac{(M_D + \omega c_\omega) r^2 \cos(\varphi)}{l((m_H + m_L) r^2 + J_R - m_H r^2 \cos^2(\varphi))}$$

$$- \frac{(l r m_H (g - \omega v) \sin(\varphi) + k_\omega v) r \cos(\varphi)}{l((m_H + m_L) r^2 + J_R - m_H r^2 \cos^2(\varphi))}$$

$$+ \frac{(M_D + \omega c_\omega) r - c_v v r^2 - c_\omega v}{(m_H + m_L) r^2 + J_R - m_H r^2 \cos^2(\varphi)}$$

$$\frac{di}{dt} = \frac{1}{L_M} \left(u_M - R_M i - k_M \left(\frac{v}{r} - \omega \right) \right)$$





Mathematical Modelling

Model parameters of the selfbalancing robot

Parameter	Value	Unit
Body (Pendulum-)length l	0.147	m
Wheel radius r	0.056	m
Upper mass m_H	1.1	kg
Lower mass m_L	0.7	kg
Moment of inertia wheels J_R	$2.2 \cdot 10^{-3}$	kg·m ²
Torque coefficient motor k_M	0.167	Nm/A
Armature resistance motor R_M	1.26	Ω
Armature inductance motor L_M	25	mH
Rotational friction coefficient c_ω	$2.6 \cdot 10^{-3}$	Nms/rad
Rolling friction coefficient c_v	2	Ns/m



Mathematical Modelling

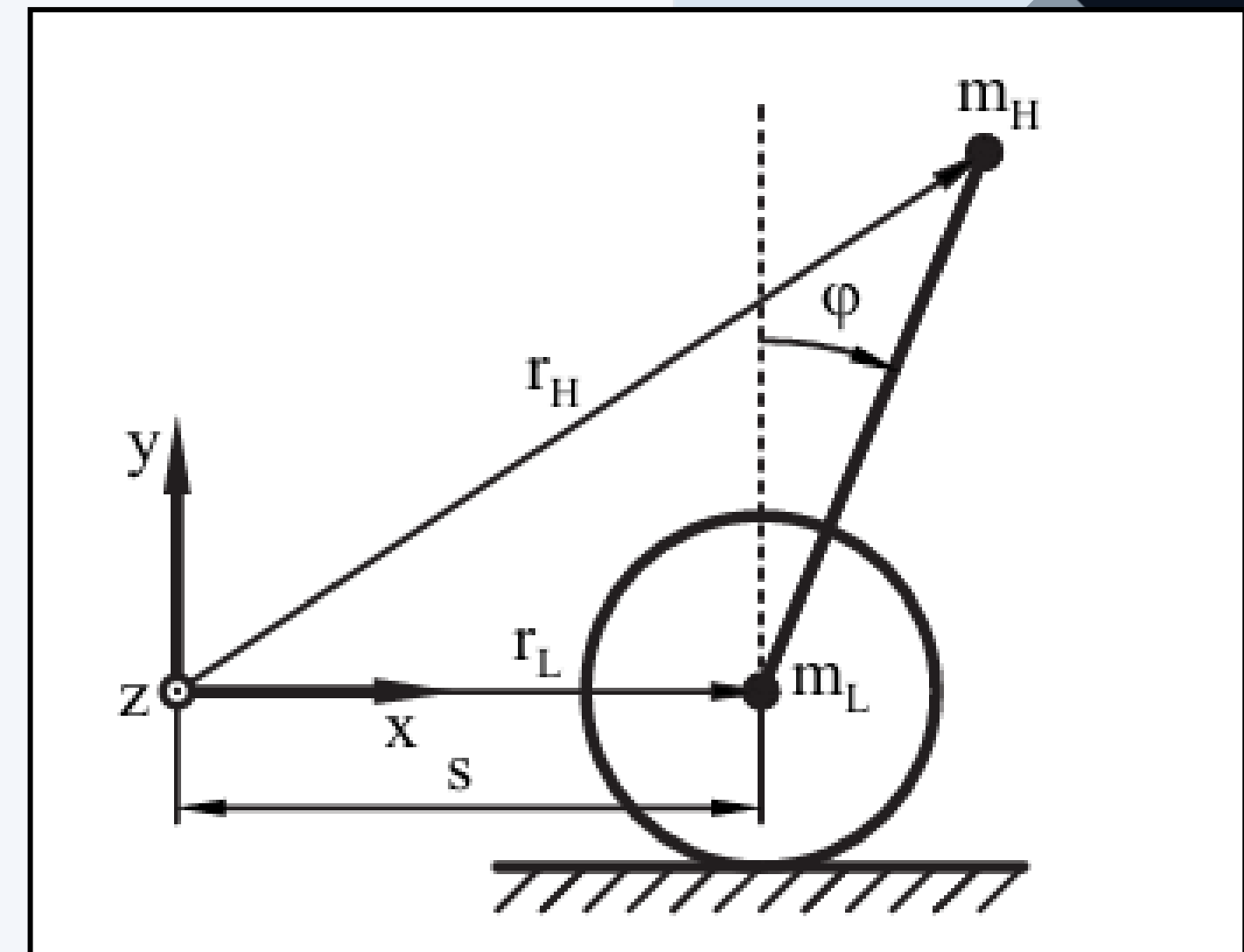
Using jacobian linearization of non linear systems about equilibrium points and taylor expansion of multivariable polynomial and putting the values of the constants, we can write

$$\dot{x} = Ax + Bu, \text{ where, } x = \begin{bmatrix} \varphi \\ \omega \\ v \\ i \end{bmatrix} \text{ given that at equilibrium point,}$$

$$\varphi_0 = \omega_0 = v_0 = i_0 = u_{M,0} = 0$$

From system parameters we get

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 119.1 & -0.4206 & 17.22 & -27.01 \\ -7.699 & 0.0457 & -2.244 & 2.938 \\ 0 & 6.68 & -119.3 & -50.4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{L_m} \end{bmatrix}$$





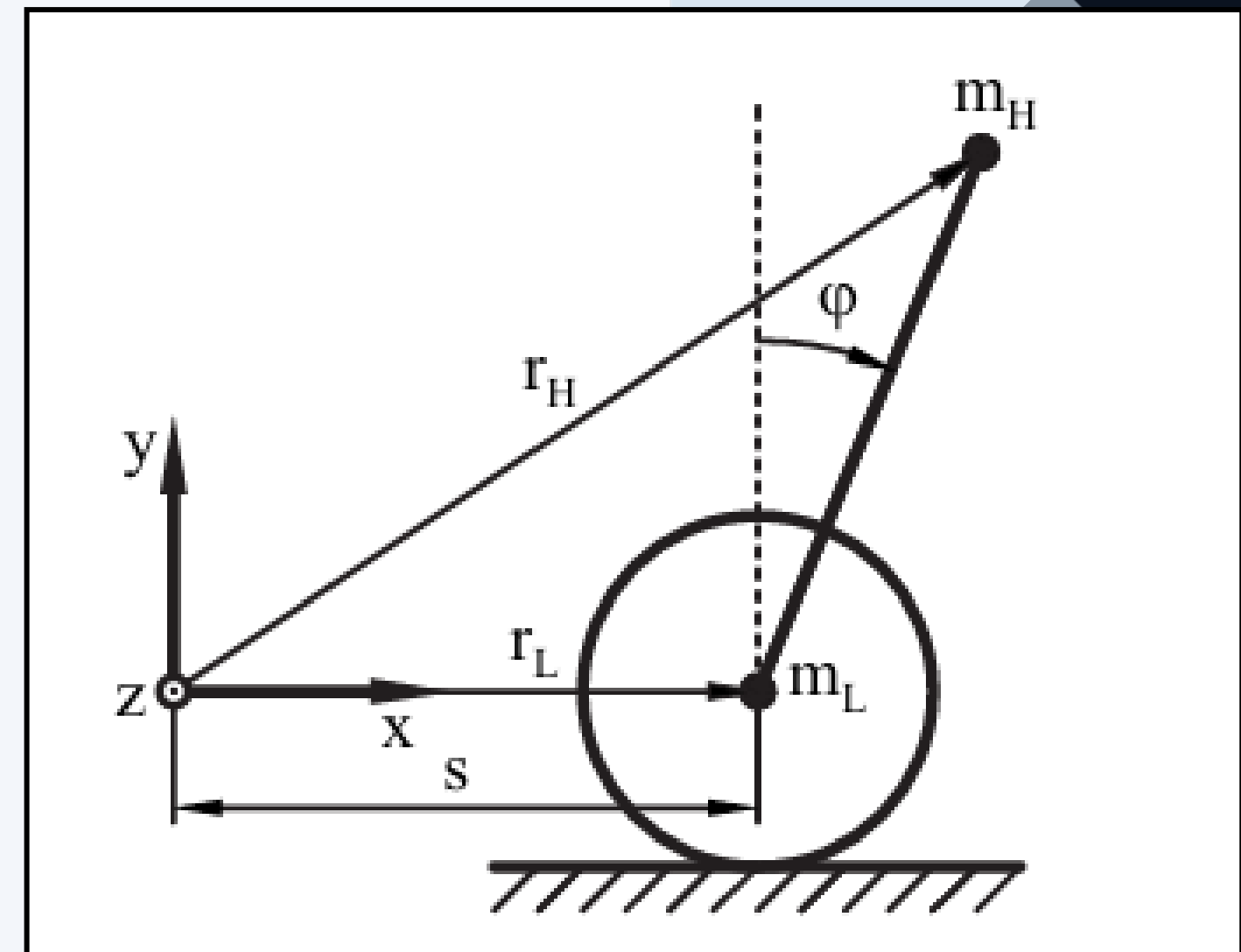
Mathematical Modelling

Mahler et al. [1] applied a full-state feedback controller using the linearized state-space model. The control law is given by:

$$u = r - \mathbf{k}^T \mathbf{x} = r - (k_\varphi \varphi + k_\omega \omega + k_v v + k_i i)$$

The feedback gain vector $\mathbf{k}$ was calculated using Ackermann's formula. The resulting gains were:

$$k_\varphi = -15.3; \quad k_\omega = -1.49; \quad k_v = -6.33; \quad k_i = 0.4$$



System modelling

(without controller)



From linearized mathematical model,

$$\dot{x} = Ax + Bu$$

$$\Rightarrow sX(s) = AX(s) + BU(s) \quad [\text{Taking Laplacian}]$$

$$\Rightarrow X(s) = s^{-1} \{AX(s) + BU(s)\}$$

From full -state feedback controller

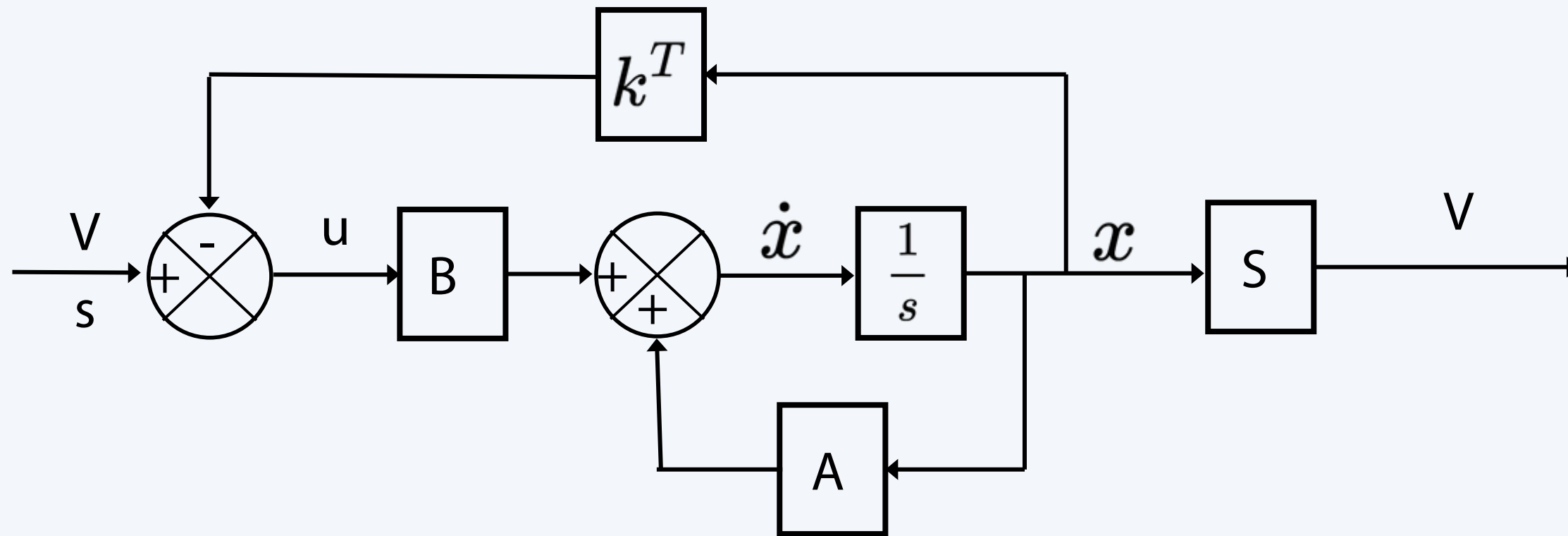
$$u = r - k^T x$$

$$\Rightarrow U(s) = R(s) - k^T X(s) \quad [\text{Taking Laplacian}]$$

System modelling

(without controller)

Block Diagram of the control system without controller:



Here , $x = \begin{bmatrix} \varphi \\ \omega \\ v \\ i \end{bmatrix}$; $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 119.1 & -0.4206 & 17.22 & -27.01 \\ -7.699 & 0.0457 & -2.244 & 2.938 \\ 0 & 6.68 & -119.3 & -50.4 \end{bmatrix}$; $B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{L_m} \end{bmatrix}$



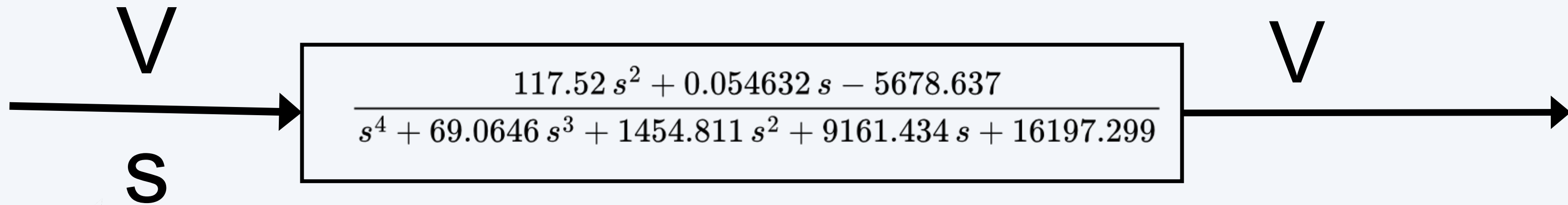
System modelling

(without controller)

$$k^T = [-15.3 \quad -1.49 \quad -6.33 \quad 0.4]$$

$$S = \text{selector matrix} = [0 \quad 0 \quad 1 \quad 0]$$

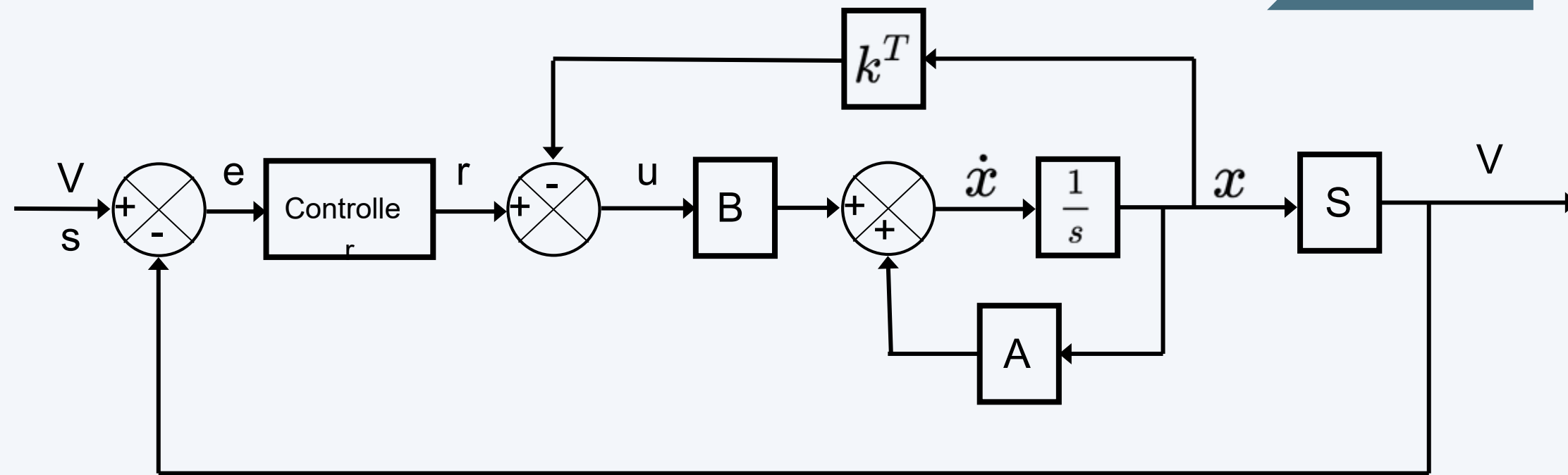
Reduced block diagram



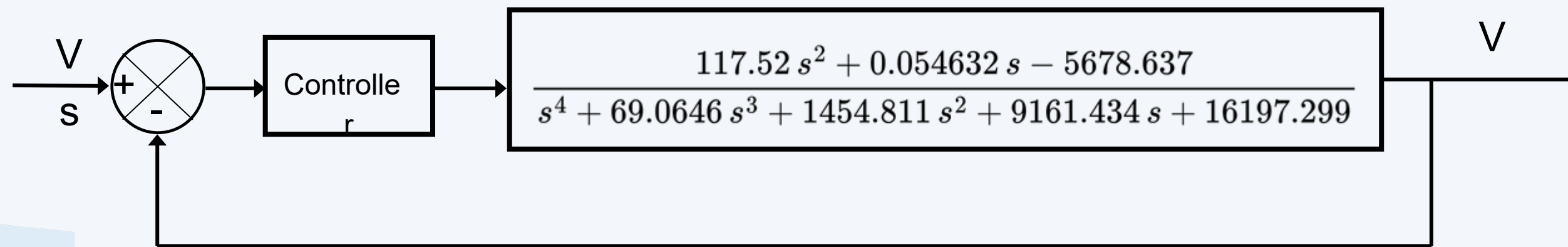
System modelling

(with controller)

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reduced block diagram with controller:





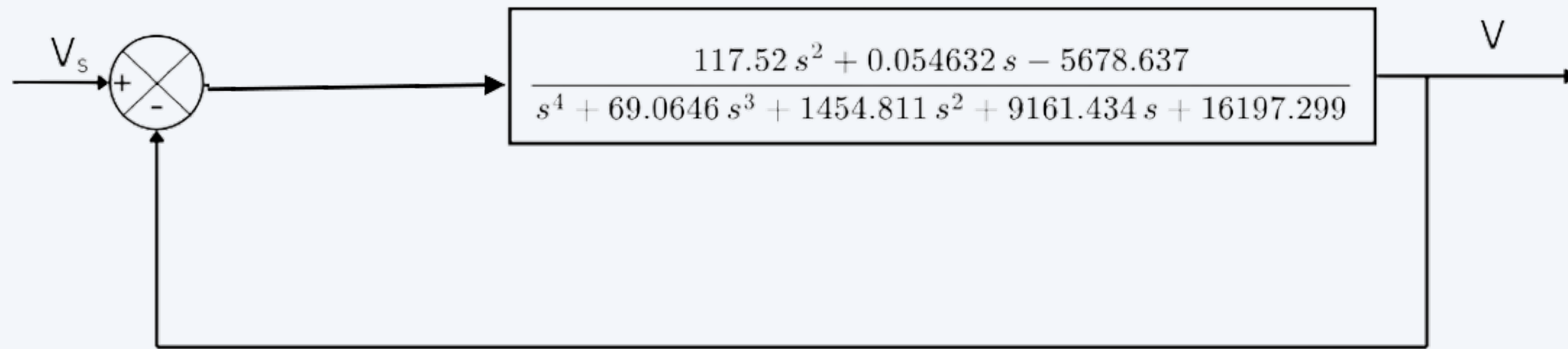
System response and Analysis (without controller)

- Stability Analysis
- Steady State Error Analysis
- System response for Unit Step Input , Ramp Input, Parabolic Input, Sinusoidal Input
- Bode Plot
- Polar Plot

Stability Analysis (without Controller)



Block Diagram of the system (without Controller) :



$$\text{Hear, } G(s) = \frac{117.52 S^2 + 0.054632 S - 5678.637}{S^4 + 69.0646 S^3 + 1454.811 S^2 + 9161.434 S + 16197.299}$$

Stability Analysis (without Controller)



Closed Loop Transfer Function :

$$G_e(s) = \frac{117.52 S^2 + 0.054632 S - 5678.637}{S^4 + 69.0646 S^3 + 1572.331 S^2 + 1961.488 S + 10518.662}$$

The characteristics equation will be :

$$S^4 + 69.0646 S^3 + 1572.331 S^2 + 1961.488 S + 10518.662 = 0$$

Stability Analysis

(without Controller)



Routh Table

s^4	1	1572.331	10518.662
s^3	69.0646	1961.488	0
s^2	1543.93	10518.662	0
s^1	1490.95	0	0
s^0	10518.662	0	0

There is no sign change at this table. So, the system is stable

Steady State Error Analysis

(without Controller)



- For Unit Step Input

$$\begin{aligned} K_p &= \lim_{s \rightarrow 0} \dot{G}(s) = \lim_{s \rightarrow 0} \frac{117.52 s^2 + 0.054632 s - 5678.637}{s^4 + 69.06466 s^3 + 1454.811 s^2 + 9161.434 s + 16197.299} \\ &= \frac{-5678.637}{16197.299} \\ &= -0.35 \end{aligned}$$

$$e_{s.s}(\text{unit step input}) = \frac{1}{1+K_p} = \frac{1}{1-0.35} = \frac{20}{13}$$

Steady State Error Analysis

(without Controller)



- For Ramp Input :

$$K_v = \lim_{S \rightarrow 0} S G(s) = 0$$

$$e_{s.s}(\text{ramp input}) = \frac{1}{K_v} = \frac{1}{0} = \infty$$

- For Parabolic Input

⋮

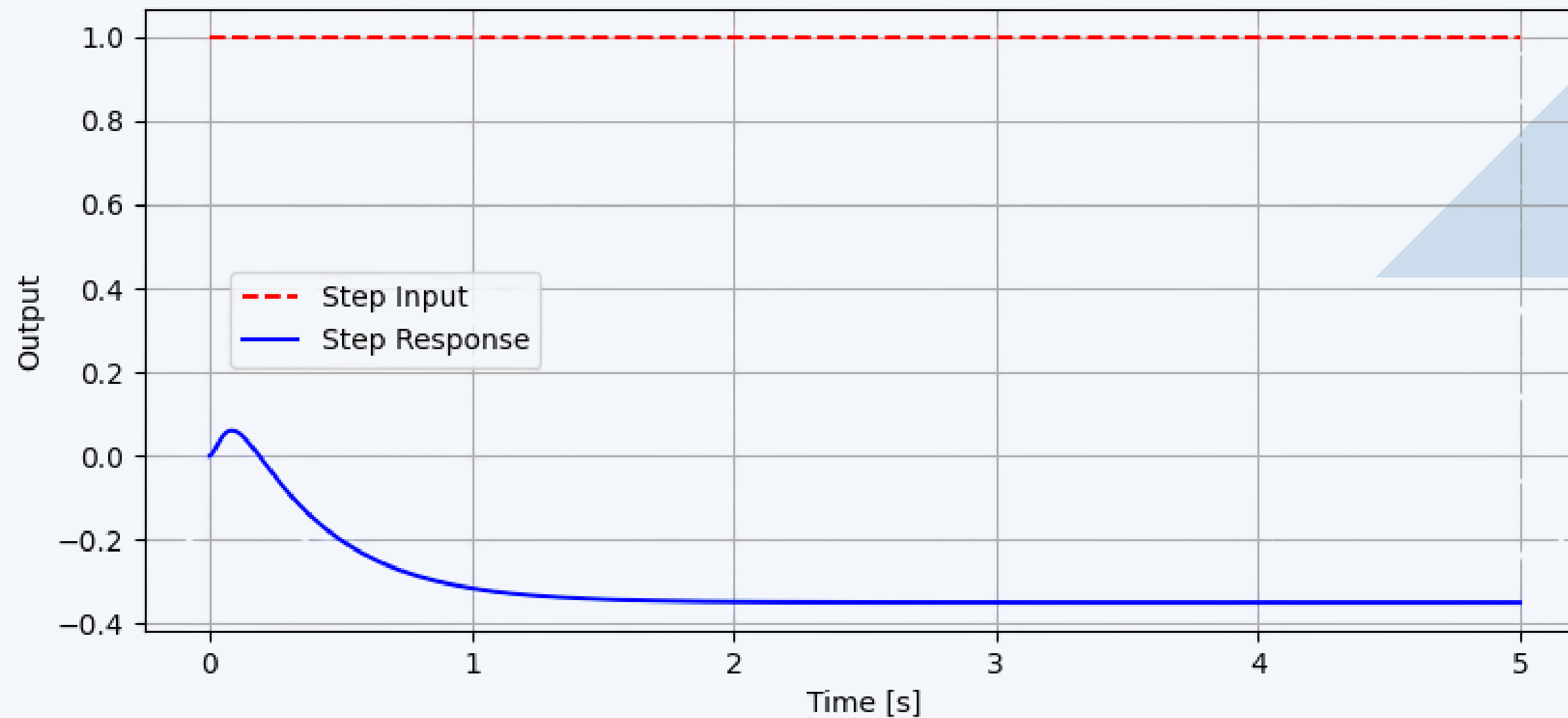
$$K_a = \lim_{S \rightarrow 0} S^2 G(s) = 0$$

$$e_{s.s}(\text{parabolic input}) = \frac{1}{K_v} = \frac{1}{0} = \infty$$

System Responses (without Controller)



Unit Step Input

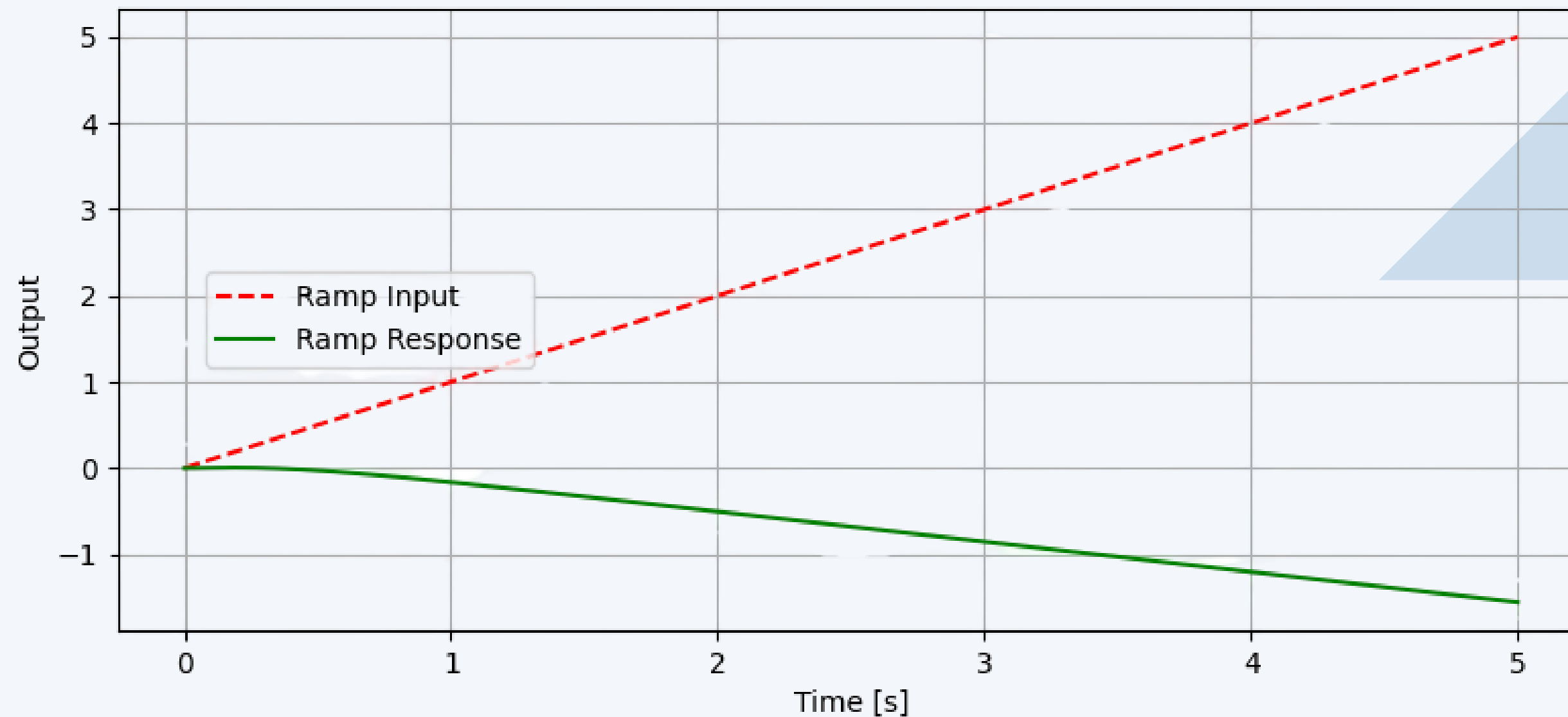


Step Input and Output

System Responses (without Controller)



Ramp Input



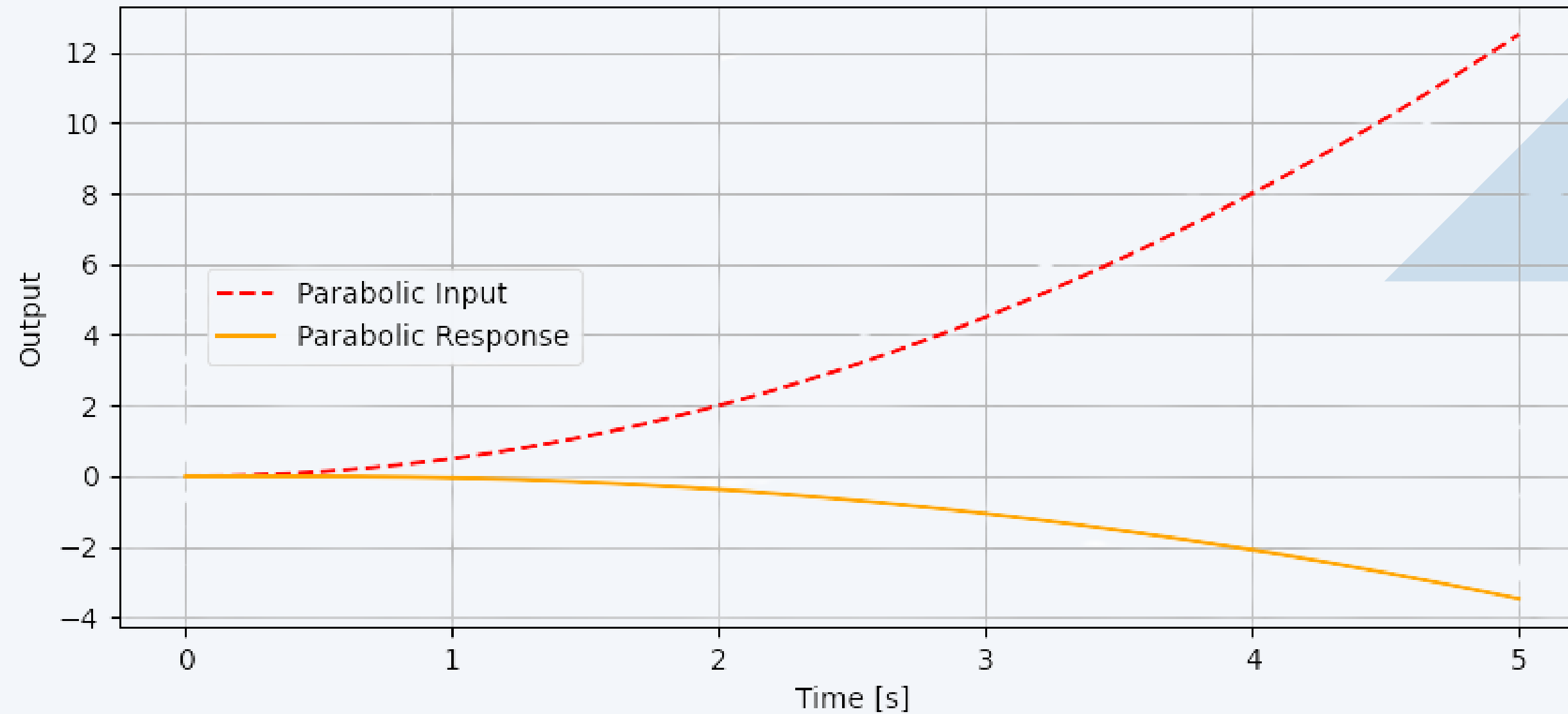
Ramp Input and Output

System Responses (without Controller)

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Parabolic Input

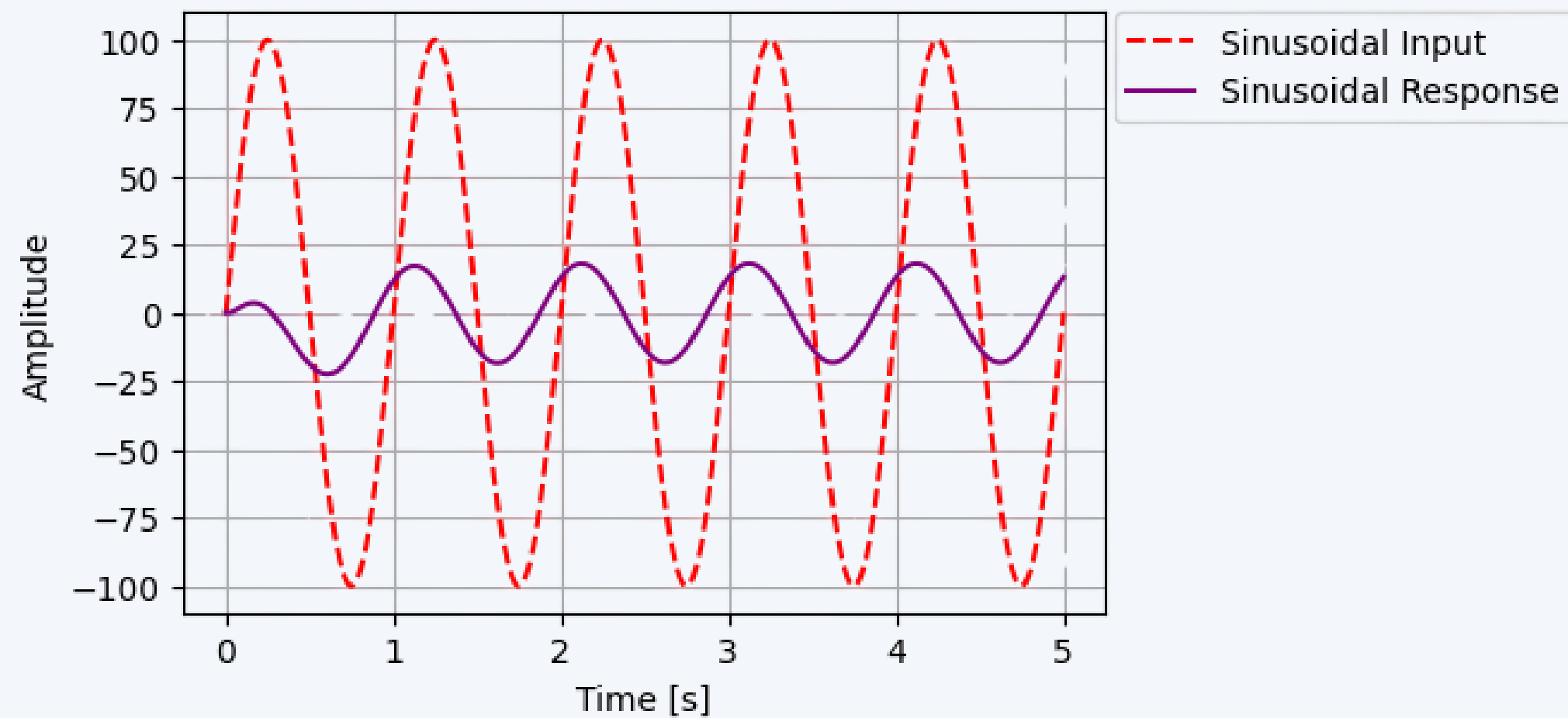


Parabolic Input and Output

System Responses (without Controller)



Sinusoidal Input



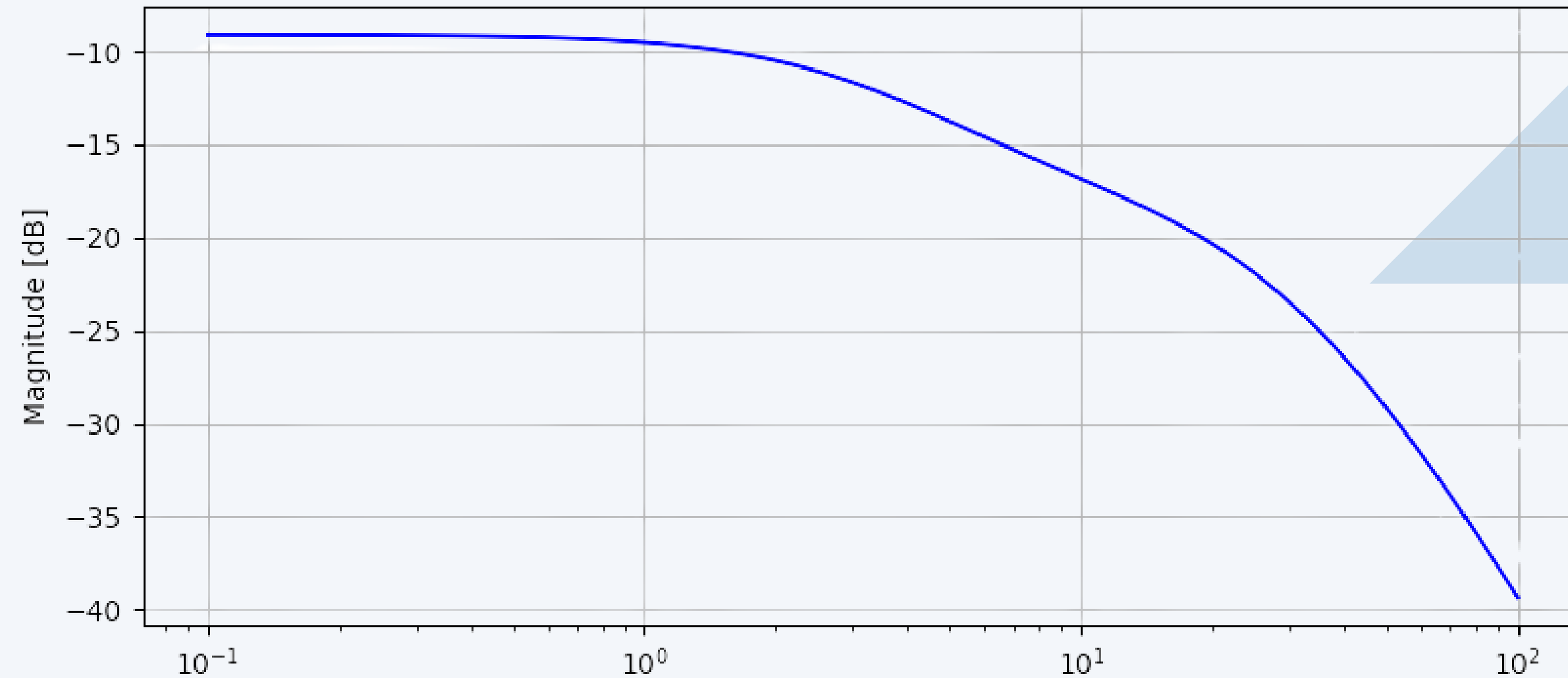
Sinusoidal Input and Output



Bode Plot

(without Controller)

Magnitude



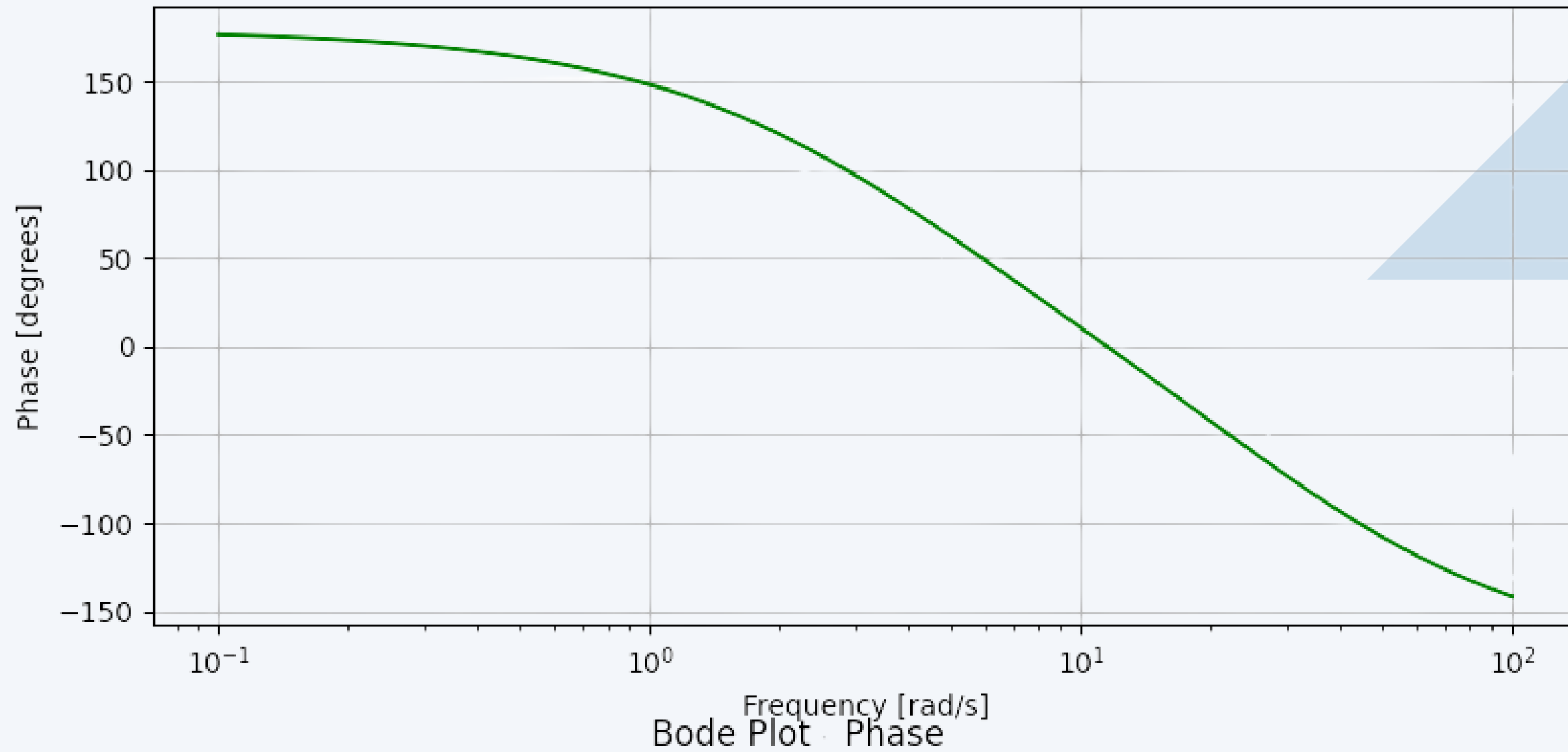
Frequency [rad/s]
Bode Plot Magnitude



Bode Plot

(without Controller)

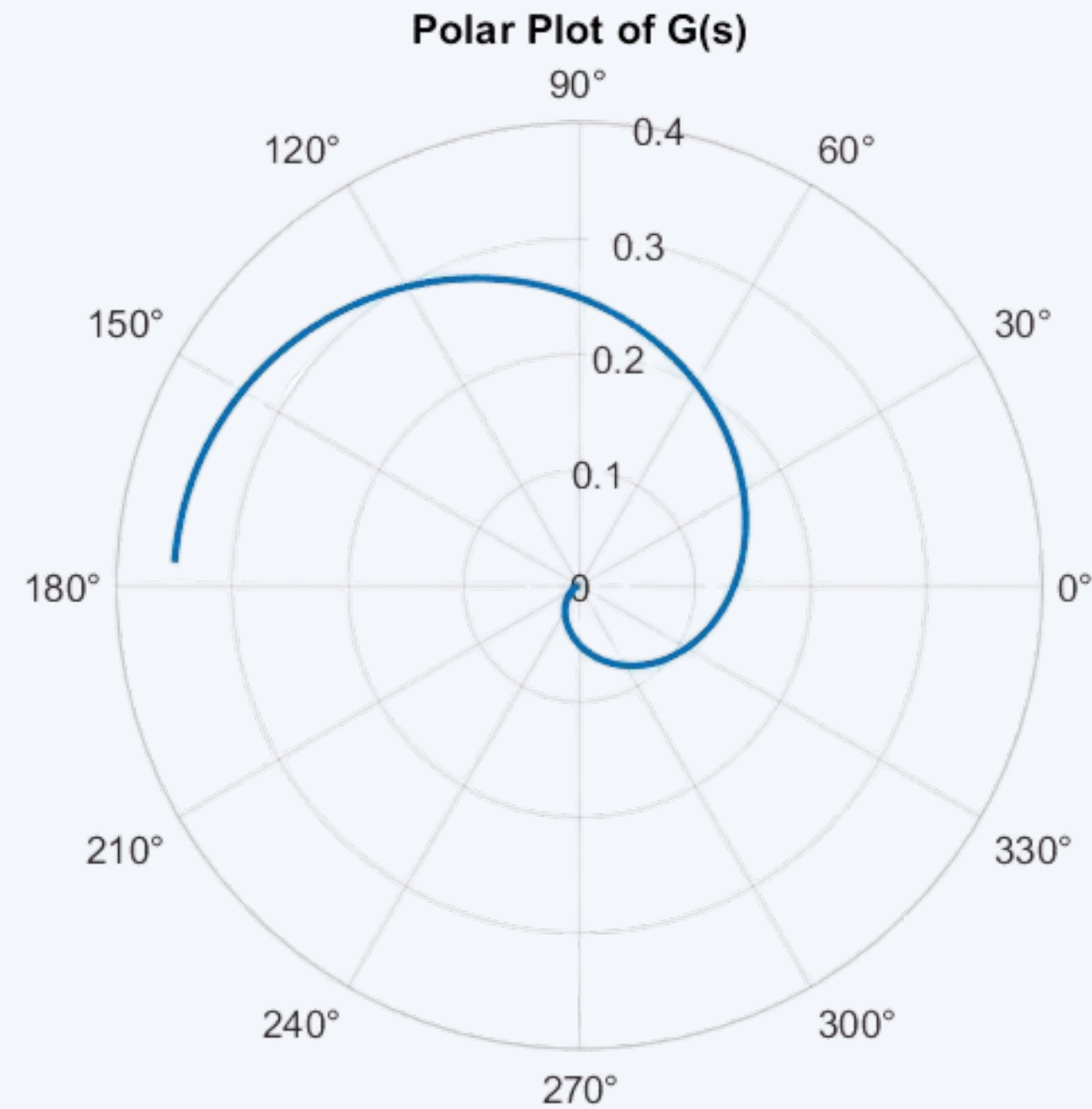
Phase



Polar Plot

(without Controller)

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PID Tuning

Using Ziegler –Nichols tuning rules



With $K_I = 0$ and $K_D = 0$ the characteristic equation of the closed-loop transfer function

$$s^4 + 69.0646s^3 + (117.5K_P + 1454.8)s^2 \\ + (0.05463K_P + 9161.434)s - (5678.637K_P - 16197.3) = 0$$

solve for marginal stability,

$$K_P = K_{cr} = -7.48486$$

$$\omega_{cr} = 11.51713 \text{ rad/sec}$$

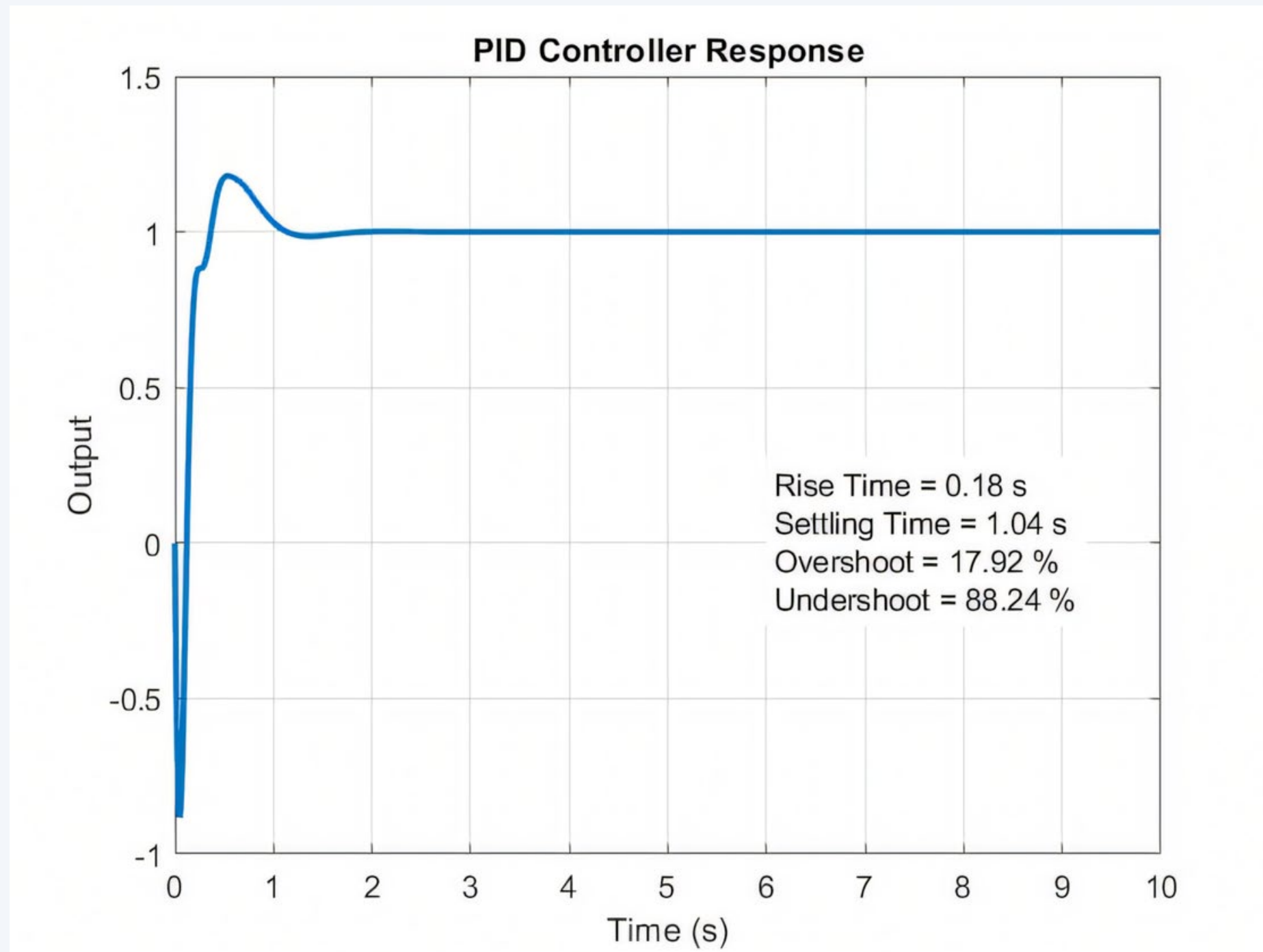
$$P_{cr} = \frac{2\pi}{\omega_{cr}} = 0.54555 \text{ sec}$$

Ziegler-Nichols Tuning Rule Based on Critical Gain K_{cr} and Critical Period P_{cr}

Type of Controller	K_P	T_i	T_d
P	$0.5K_{cr}$	∞	0
PI	$0.45K_{cr}$	$\frac{1}{1.2}P_{cr}$	0
PID	$0.6K_{cr}$	$0.5P_{cr}$	$0.125P_{cr}$

PID Tuning Results

Using Ziegler –Nichols tuning rules



Parameter	P	PI	PID
K_P	-3.7424	-3.3682	-4.4909
K_I	0	-7.4087	-16.4638
K_D	0	0	-0.3063
Performance Metrics			
Rise Time (s)	0.10	0.20	0.18
Settling Time (s)	0.98	1.44	1.04
Overshoot (%)	33.45	7.65	17.92
Undershoot (%)	26.97	27.26	88.24



Summary of Recent Research using different algorithms

Paper Reference	PID Tuning Algorithm Used	Comments and Findings
Kumar et al. (2012) - "Tuning of PID Controller of Inverted Pendulum Using Genetic Algorithm"	Genetic Algorithm (GA)	Similar to our work as it uses GA-based PID tuning on an inverted pendulum system, which is analogous to a two-wheeled self-balancing robot.
Olmez et al. (2022) - "Clonal selection algorithm-based control for two-wheeled self-balancing mobile robot"	Clonal Selection Algorithm (CSA), PSO, ABC, DE, GoldSall, Cuckoo Search	Directly relevant as it involves a two-wheeled self-balancing robot and demonstrates the comparative effectiveness of metaheuristic optimization methods for PID tuning.
Hassanzadeh & Mobayen (2011)- "Controller Design for Rotary Inverted Pendulum System Using Evolutionary Algorithms"	GA, PSO, ACO	Relevant as it tests multiple PID tuning methods, giving insights into algorithm effectiveness for balancing tasks similar to our two-wheeled robot scenario.
Yetayew & Tesfaye (2024) "A Comparative Analysis of Metaheuristic Algorithms Tuned Supper Twisting Sliding Mode Control of a Self-balancing Segway"	GWO, GA, PSO	Highly relevant; directly involves a two-wheeled self-balancing Segway system and compares multiple PID tuning algorithms, closely aligned with our project's goals.

PID Tuning

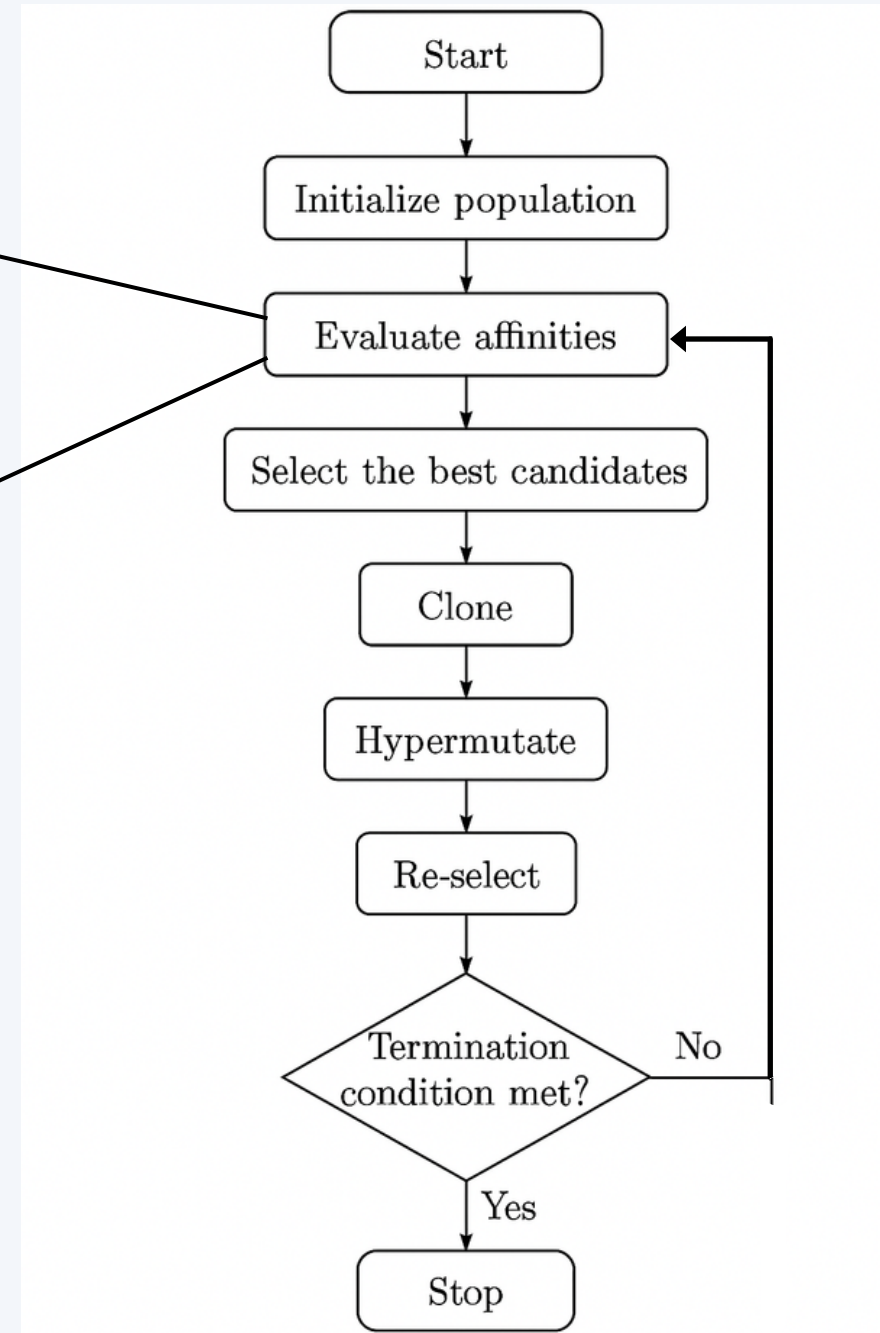
Using CSA (Clonal Selection Algorithm)

Integral Squared Error (ISE) cost function

$$ISE = \int_0^{\infty} e(t)^2 dt$$

Integral of Time-weighted Absolute Error (ITAE)

$$ITAE = \int_0^{\infty} t|e(t)|dt$$



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Core Parameters for Clonal Selection Algorithm

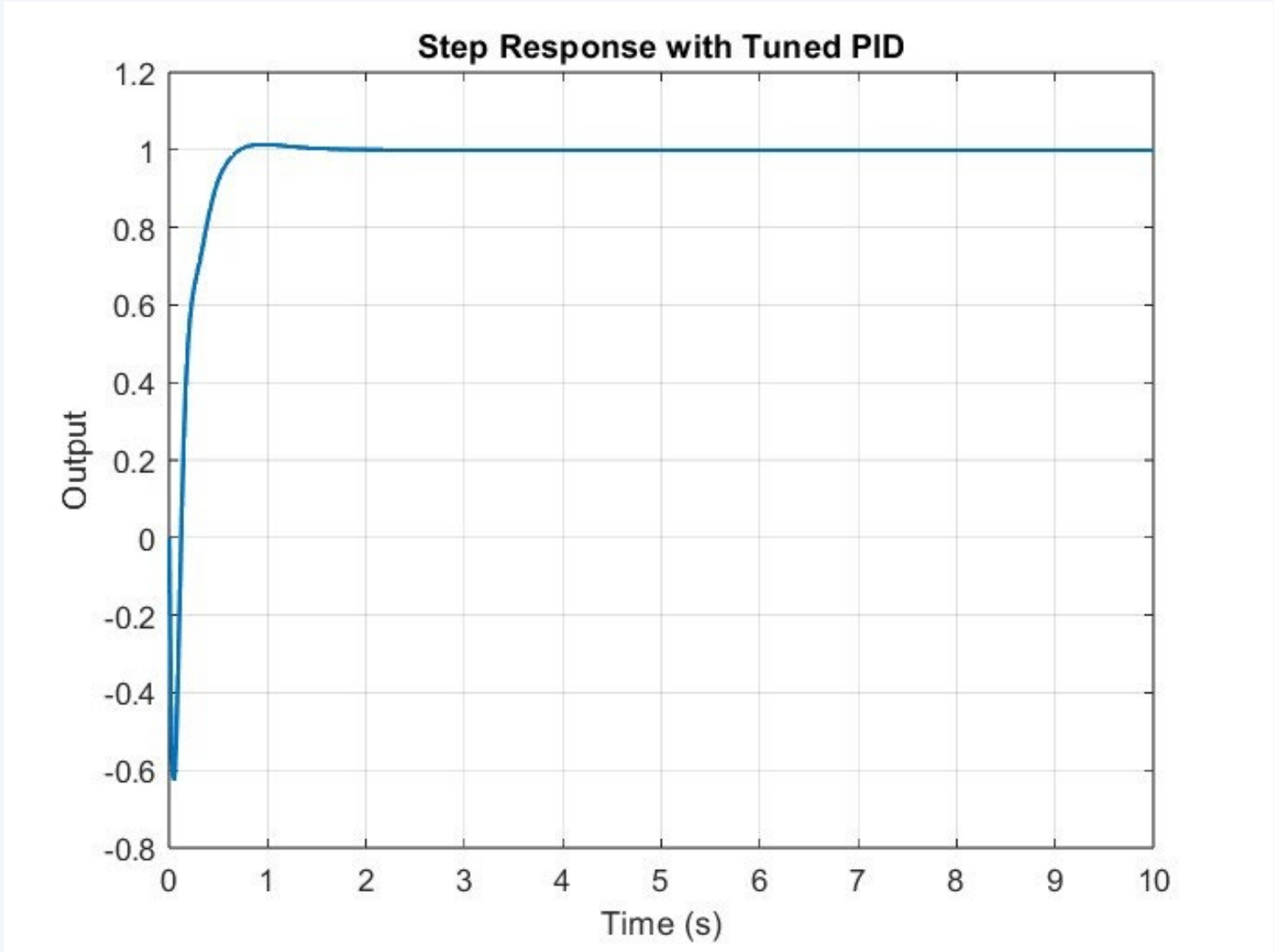
Parameter	Value
Population Size	30
Cloning Factor	5
No. of Selected Clones	5
No. of Random Inserts	2
Base Mutation Rate	0.2
Maximum Generations	100

PID Tuning Results

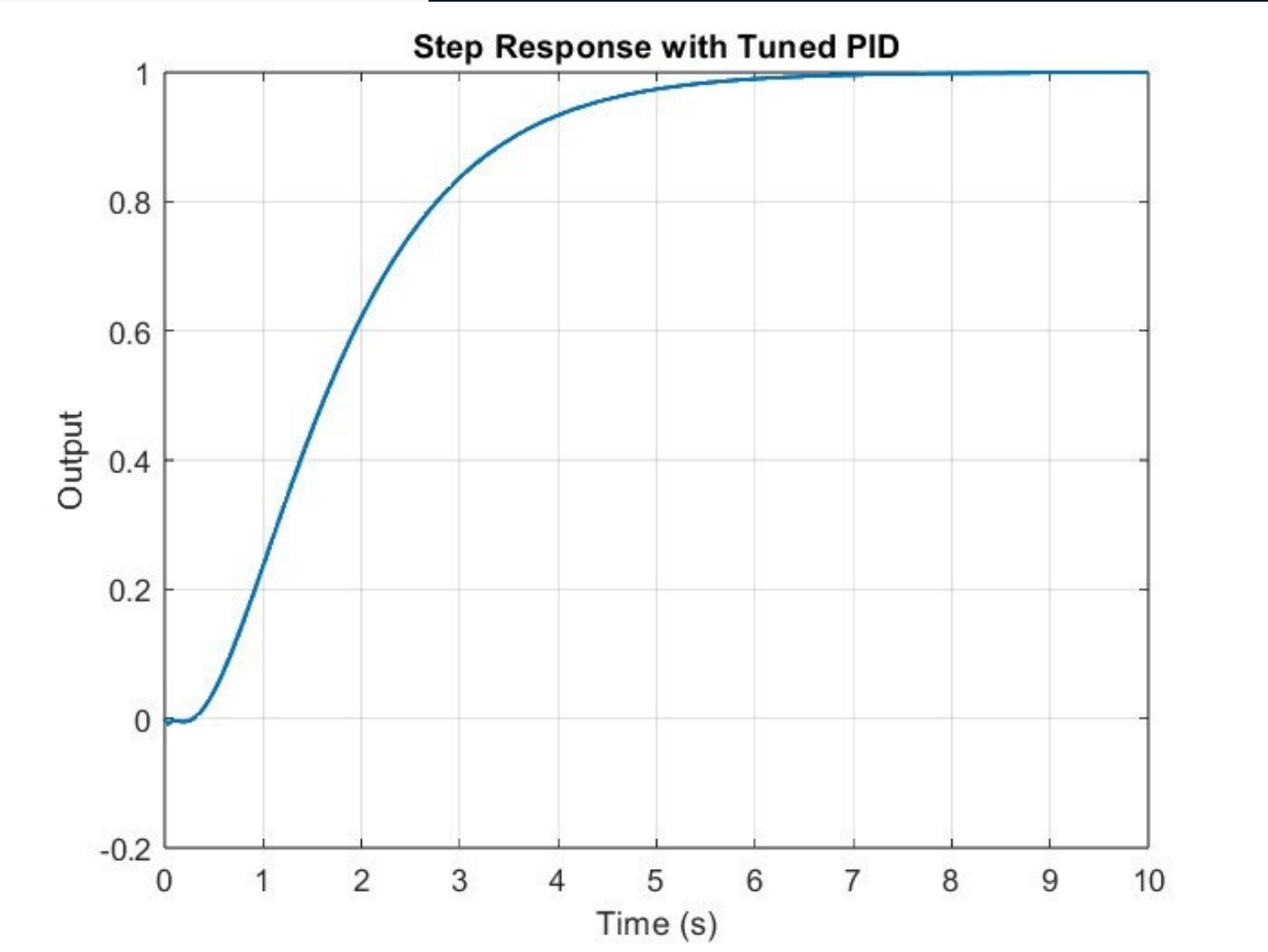
Using CSA (Clonal Selection Algorithm)

Step response with Tuned PID controller

Parameter	Standard ISE (with Undershoot)	Penalized ITAE (Zero Undershoot)
Optimal K_P	-3.773710	0.011320
Optimal K_I	-9.662423	-1.481346
Optimal K_D	-0.246369	-0.006500
Performance Metrics		
Rise Time (s)	0.3383	2.8752
Settling Time (s)	0.6286	5.2749
Overshoot (%)	1.37	0.00
Undershoot (%)	62.52	0.78

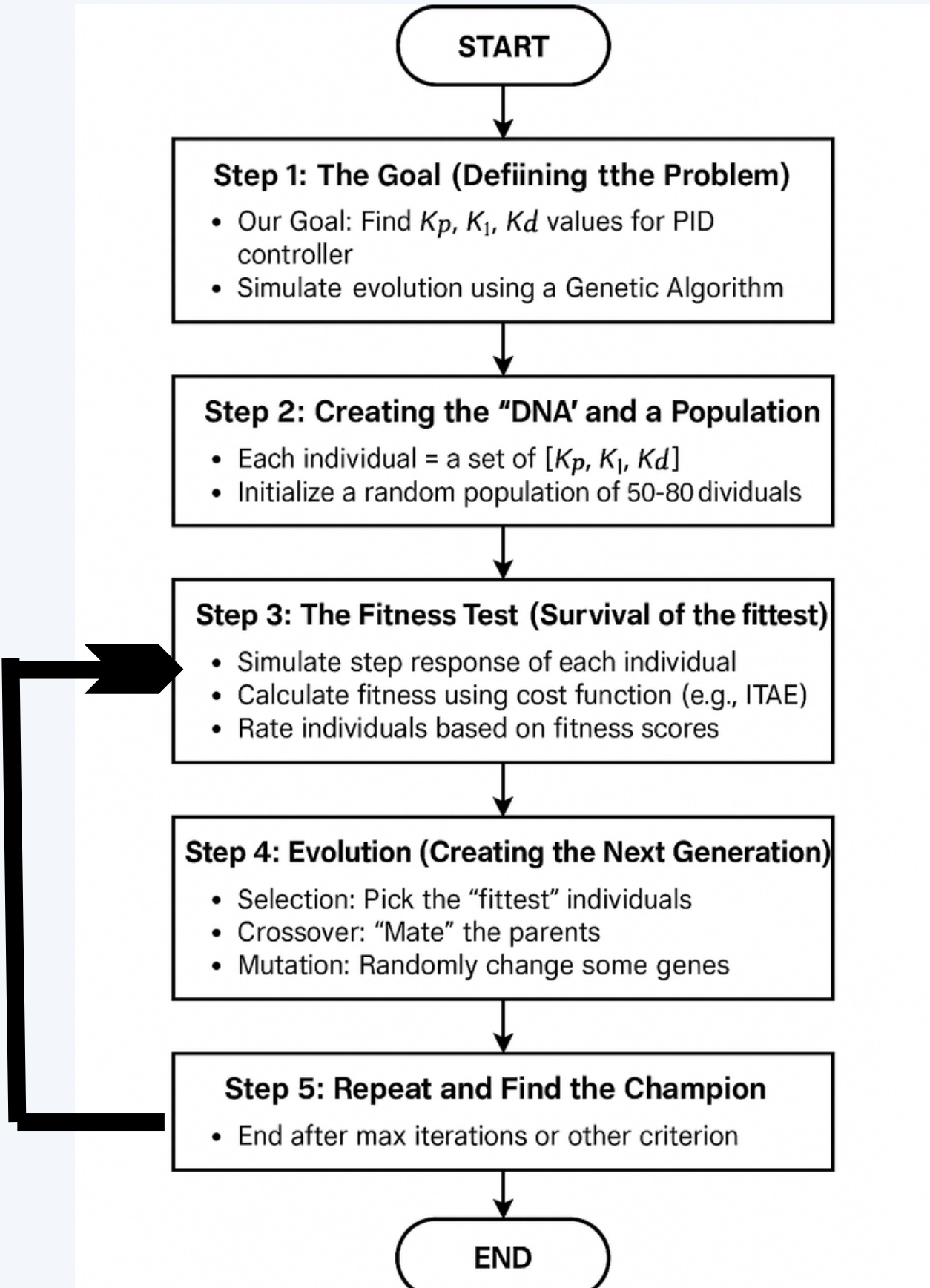


Allowing undershoot



Penalized Undershoot with cost function

Genetic Algorithm



Parameters and Values

Generations	100
Population Size	80
Crossover Rate	0.8
Mutation Rate	0.1

Cost Function

Integral of Time-weighted Absolute Error (ITAE)

$$ITAE = \int_0^{\infty} t|e(t)|dt$$

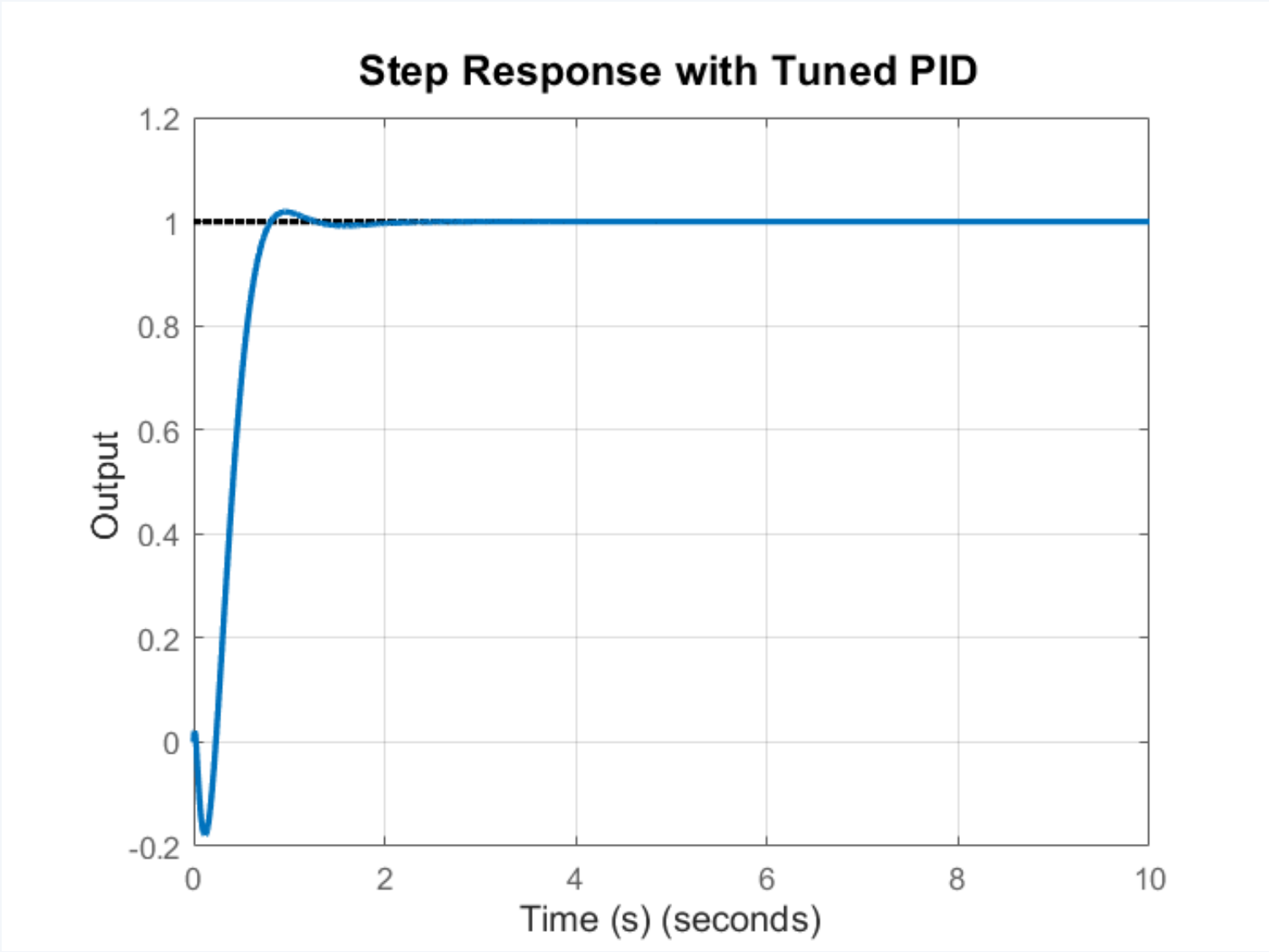


PID Tuning Results

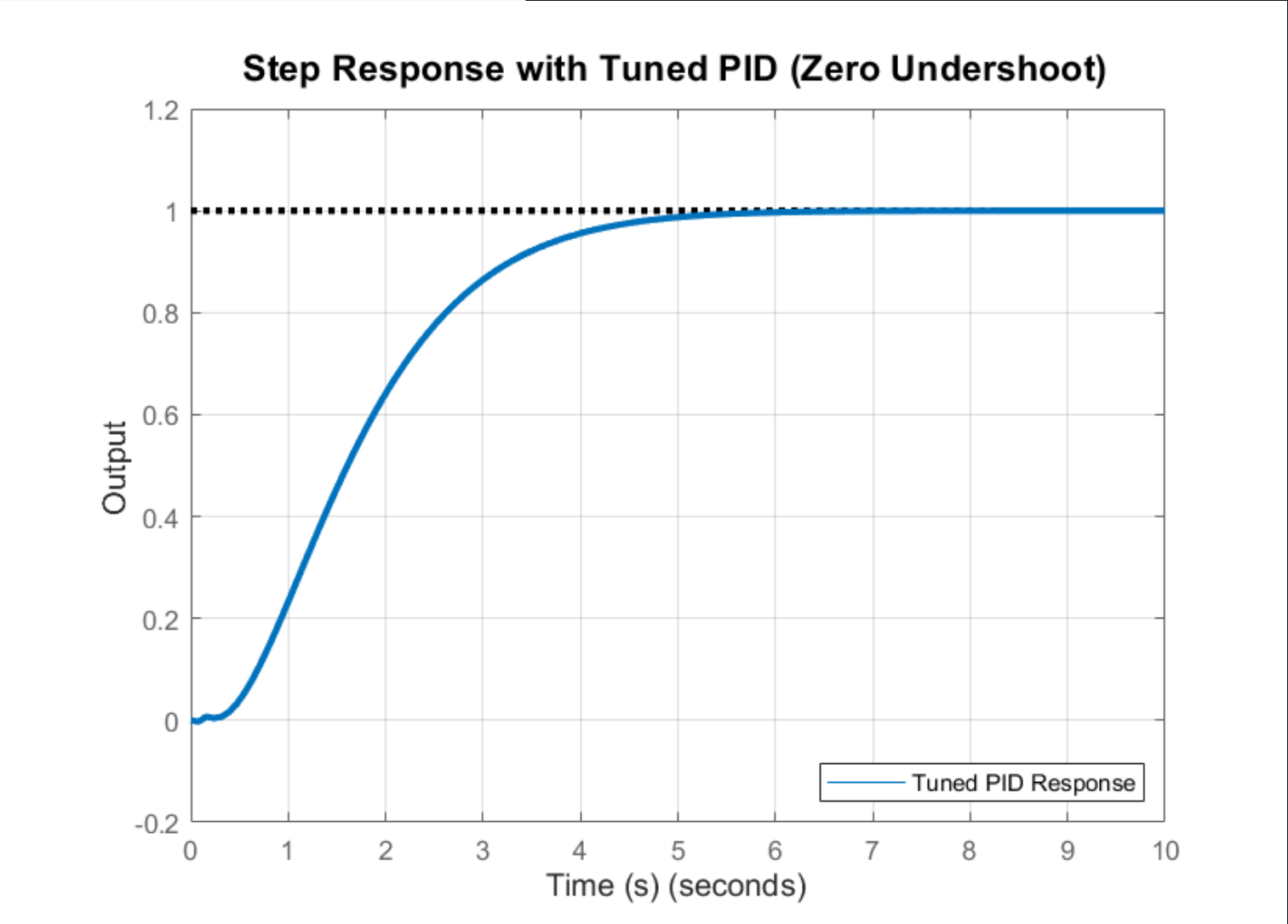
Using GA (Genetic Algorithm)

Step response with Tuned PID controller

Parameter	Standard ITAE	ITAE + Undershoot Penalty
Optimal K_P	-2.1769	0.0651
Optimal K_I	-6.2132	-1.5575
Optimal K_D	0.0385	-0.0200
Performance Metrics		
Rise Time (s)	0.3722	2.5945
Settling Time (s)	0.7537	4.6508
Overshoot (%)	1.8503	0.0000
Undershoot (%)	17.9021	0.2404



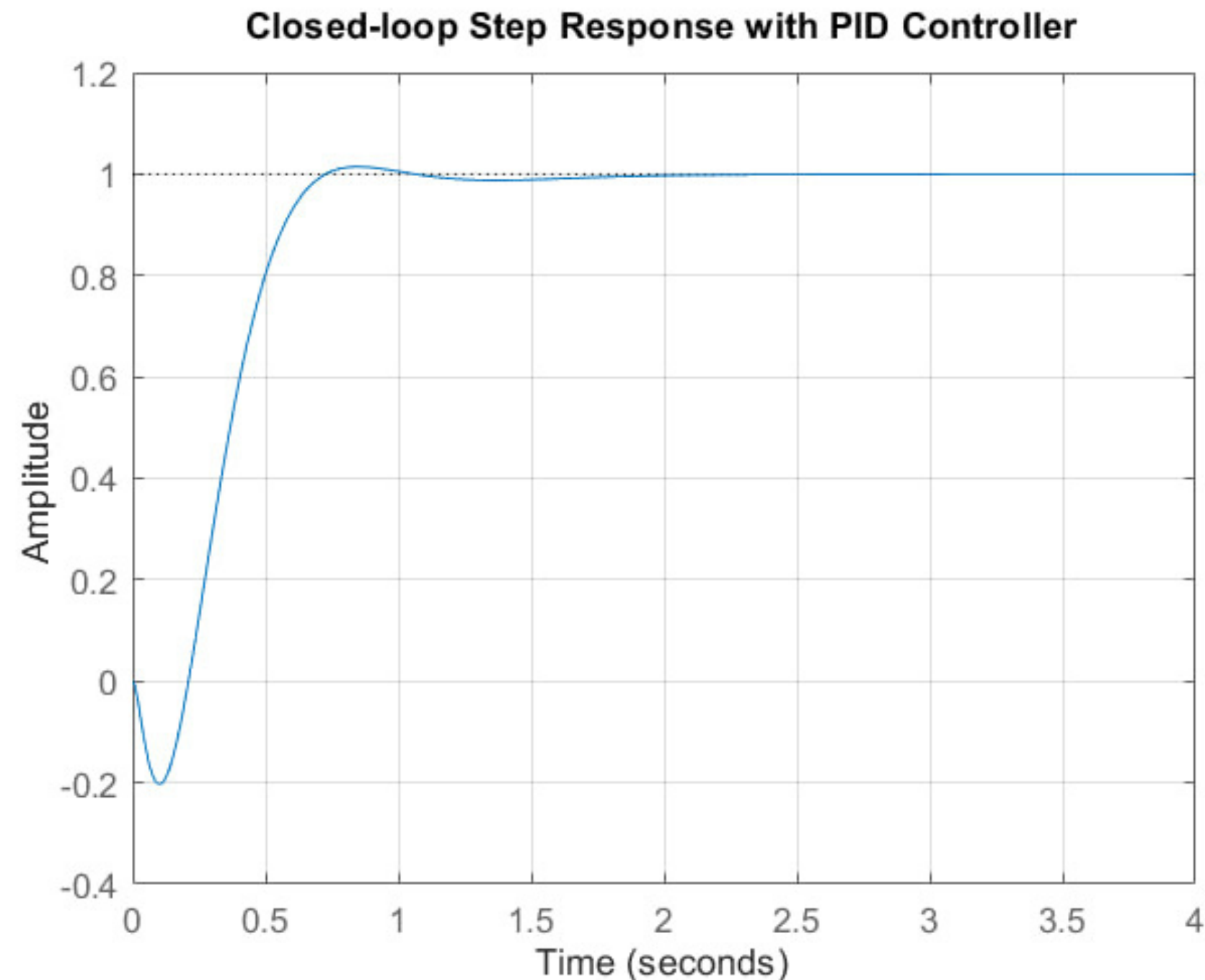
Allowing undershoot



Penalized Undershoot with cost function

PID Tuning Results

Using MATLAB Auto -Tuner Software

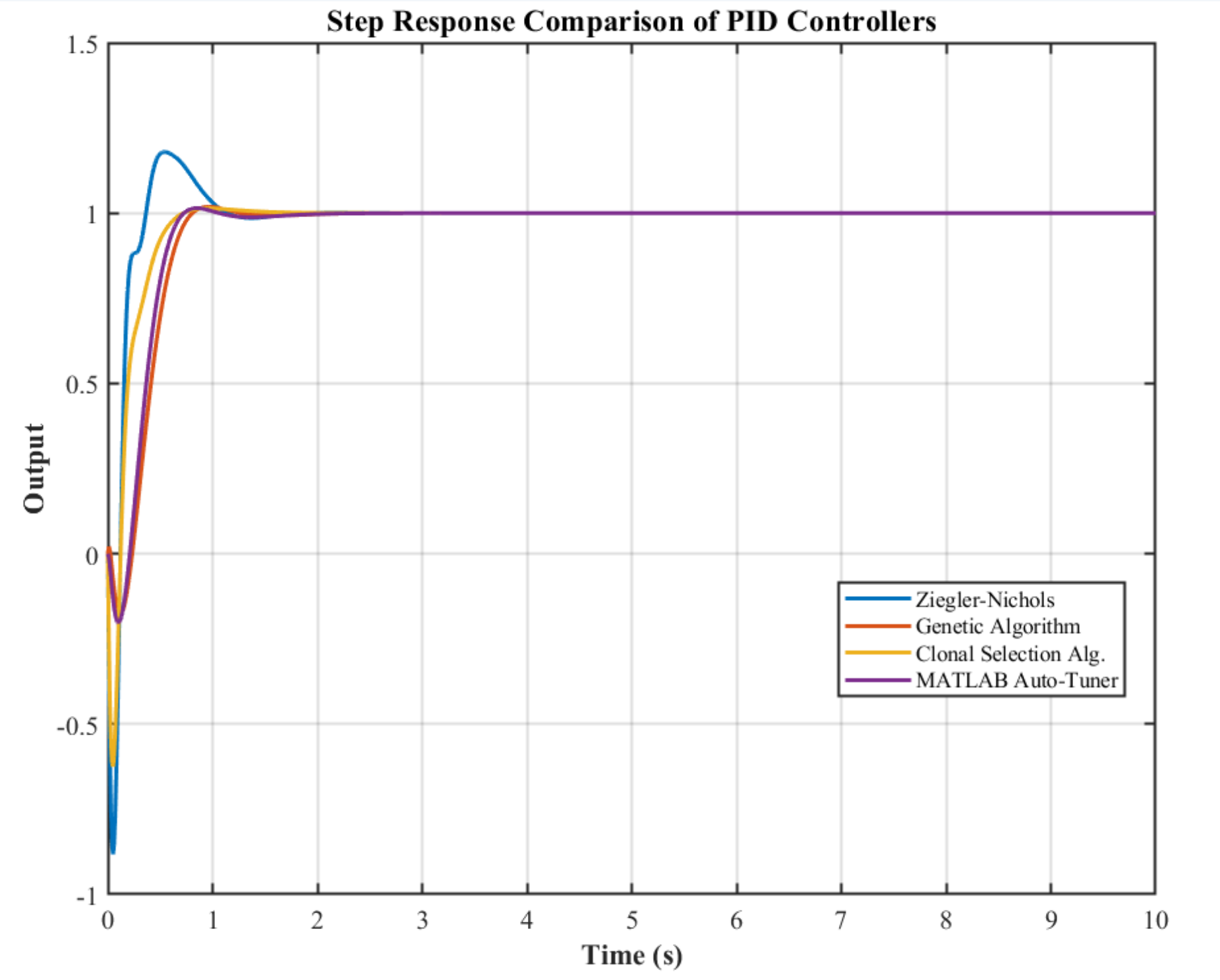


Parameter / Metric	P-Tuned	PI-Tuned	PID-Tuned
K_P	-4.95	-2.5568	-2.5568
K_I	0	-6.7867	-6.7867
K_D	0	0	0
Rise Time (s)	0.0695	0.329	0.329
Settling Time (s)	1.6864	0.674	0.674
Overshoot (%)	69.18	1.44	1.44
Undershoot (%)	38.07	20	20



Comparison of Controllers

Without Undershoot Penalization

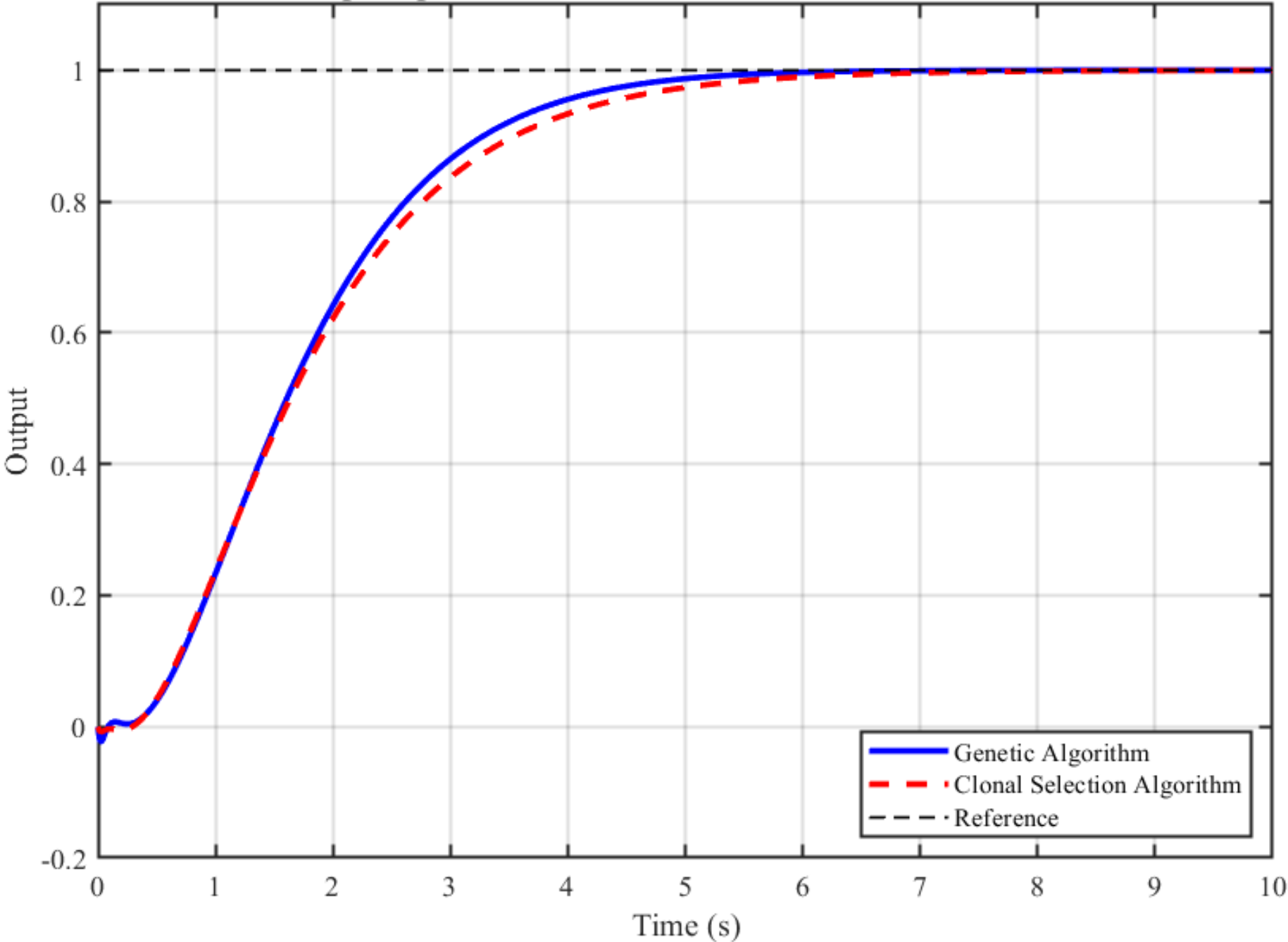


Parameter / Metric	Ziegler-Nicholas	Genetic Algorithm	Clonal Selection Alg.	MATLAB Auto-Tuner
Controller Gains				
K_p	-4.490922	-2.1769	-3.773710	-2.5568
K_i	-16.46375	-6.2132	-9.662423	-6.7867
K_d	-0.3062543278	0.0385	-0.246369	0
Performance Metrics				
Rise Time (s)	0.1773	0.3722	0.3383	0.329
Settling Time (s)	1.0384	0.7537	0.6286	0.674
Overshoot (%)	17.92	1.8503	1.37	1.44
Undershoot (%)	88.24	17.9021	62.52	20

Comparison of Controllers

With Undershoot Penalization

Step Response: Zero Undershoot Tuned PID Controllers



Parameter / Metric	Genetic Algorithm	Clonal Selection Alg.
Controller Gains		
K_p	0.0651	0.011320
K_i	-1.5575	-1.481346
K_d	-0.0200	-0.006500
Performance Metrics		
Rise Time (s)	2.5945	2.8752
Settling Time (s)	4.6508	5.2749
Overshoot (%)	0.0000	0.0000
Undershoot (%)	0.2404	0.78

Conclusion

1. PID controllers offered the best overall performance
2. Ziegler Nichols tuning produced fast responses but resulted in large overshoot and undershoot
3. MATLAB's auto tuner provided acceptable PI and PID responses but lacked fine tuning capabilities
4. Metaheuristic tuning with GA and CSA enabled precise control, with undershoot penalization improving stability and accuracy.
5. Genetic Algorithm achieved the best penalized result with lowest undershoot and faster settling, outperforming CSA.



Reference

- [1] B. Mahler, J. Haase, Balancing an inverted pendulum on a mobile robot, in: 2013 19th International Conference on Automation and Computing (ICAC), IEEE, 2013, pp. 1 –6. doi:10.1109/IConAC.2015.7313975.
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27 July 2025